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An improved parameterization method for B-spline curve and surface interpolation

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ABSTRACT

The parameterization method plays a critical role in B-spline interpolation. Some of the well-known parameterizations are the uniform, centripetal, chord length, Foley and universal methods. However, the interpolating results of these methods do not always satisfy all data features. In this study, we propose a new parameterization method which aims to improve the wiggle deviation of the interpolation, especially when interpolating the abrupt data interpolation. This new method is a refined centripetal method. The core of refinement is introducing the osculating circle at each data point. Besides the new parameterization method, we also design a fine wiggle validation method to verify the performance of all methods. In this paper, the proposed method is compared with centripetal, chord length, Foley, uniform and universal methods in both curve and surface cases. As a result, the proposed method has fewer wiggles than the centripetal method and other methods in the cases of abrupt-changing data. In addition, this refined method is stable for all kinds of data types, including free-form data distribution in this paper. The proposed method has fewer drawbacks than other methods, such as wiggles, oscillations, loops, and peaks, among others. More advantage, the proposed method is less influenced by the degree changing.
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1. Introduction

B-spline is one of the most popular methods for modeling curves and surfaces in the field of Computer Aided Geometric Design (CAGD) due to its properties of smoothness and localness. To fit a given set of points, a B-spline can be used either in approximation or in interpolation. Herein, we focus on interpolating a B-spline to a given ordered chain or grid set of points.

First consider a pth-degree B-spline curve global interpolation [1] to a sequence of given points $\{P_i\}, i = 0, \ldots, n$, where $n \geq p$ and no two consecutive points are the same. Let the B-spline curve have control points $\{B_i\}, i = 0, \ldots, n$. By assigning a parameter value or interpolating node, $\bar{u}_i$, to each P_i, and determining a knot vector $U = \{u_0, \ldots, u_m\}$ to find $\{B_i\}$, the problem can result in a $(n+1) \times (n+1)$ system of the linear equations [1]

$$P_i = C(\bar{u}_i) = \sum_{j=0}^{n} B_j N_{j,p}(\bar{u}_i) \quad i = 0, \ldots, n \tag{1}$$

where $N_{j,p}(u)$ are the normalized B-spline basis functions of degree p defined on the knot vector U. Clearly, different choice of the parameterization $\{\bar{u}_i\}$ and U directly affect the shape of the resulting curve. What we concern is to obtain a "fair" or "pleasing" curve through the data points. Generally speaking, a "fair" curve is expected to present the features of the data distribution. This problem can induce the parameterization of the data points [2]. If the parameterization is chosen appropriately, then we can do less the modification or refinement. The parameterization generally contains node choosing and knot determination. Furthermore, they influence the shape of curves in different ways. One or both of them have been the subject of many studies. In this paper, we focus on how to determine the appropriate node values beforehand, that is, to establish an appropriate a *priori* estimate of data points in order to obtain a satisfactory shape.

In B-spline interpolation, the most common parameterizations are the uniform, chord length and centripetal methods [1,3,4]. The parameterization formula of these three methods, with setting $\bar{u}_0 = 0$, can be written in the general exponent form as

$$\frac{\bar{u}_{i+1} - \bar{u}_i}{\bar{u}_i - \bar{u}_{i-1}} = \frac{\Delta_{i+1}}{\Delta_i}, \quad \Delta_i = \|L_i\|^e, \ 1 \leq i \leq n \tag{2}$$

where $\|L_i\|$ denotes the norm of the vector L_i from P_{i-1} to P_i. It can then be reduced to the uniform method with $e = 0$, to the chord length method with $e = 1$, and to the centripetal method with $e = 1/2$. Setting node parameters is most easily accomplished using the

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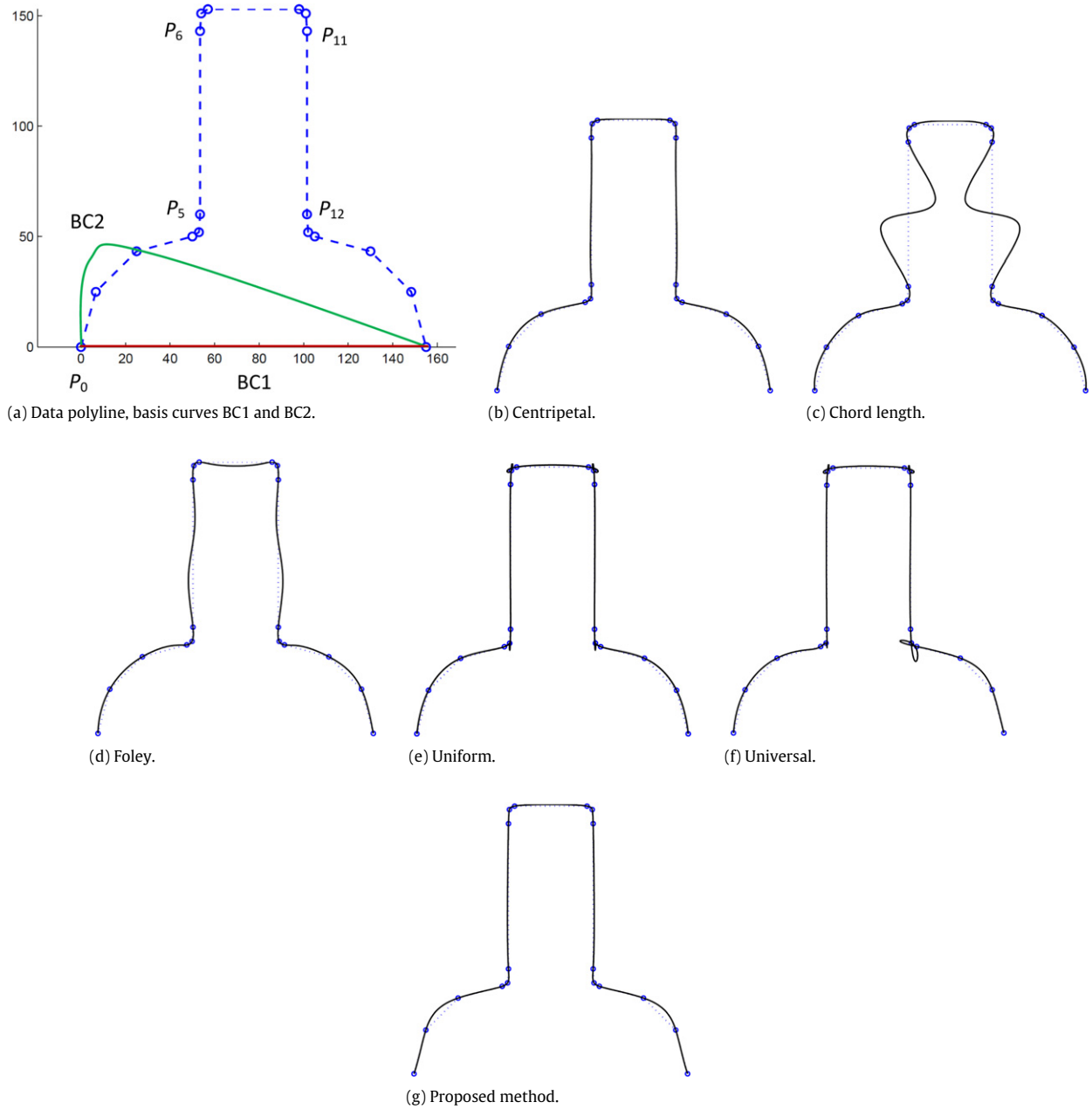


(a) Data polyline, basis curves BC1 and BC2.

(b) Centripetal.

(c) Chord length.

(d) Foley.

(e) Uniform.

(f) Universal.

(g) Proposed method.

Fig. 1. Data and cubic B-spline interpolating results using 6 different methods as applied to a symmetric example with corners considered in Lee [2].

uniform method, but this method usually dismisses the geometry of the data points [1–3,5], easily resulting in a cusp or a loop in the resulting curve, as shown in Fig. 1(e). The chord length method then generally is adopted based on an analogy to driving a car: as the car is driven at an approximately uniform speed, the node value space at each data point (seen as the differences in arrival times) is proportional to the length of each chord in data points [2,3,5]. The chord length method usually produces better results than does the uniform method, but not in all cases. The chord length cannot produce curves with corners at the data points, and often yields a "rounded" curve [3]. In other words, the chord length method easily results in a curve having bulges or one that wanders away from the data points. Moreover, the resulting curve of the chord length wiggles unpredictably between the two adjacent points as seen in Fig. 1(c).

In 1989, Lee proposed the centripetal model based on the physical heuristics of driving a car [2]. As the car smoothly passes through a data point, the method is derived from the suggestion that the normal force along the path should be proportional to the change in the angle. On average, the centripetal method produces better results than the uniform and chord length methods.

Aside from these commonly-used methods, G. Nielson and T. Foley developed another method to cope with abrupt data distribution [4,6] as

$$\Delta_i = d_i \left(1 + \frac{3\hat{\theta}_i d_{i-1}}{2(d_{i-1} + d_i)} + \frac{3\hat{\theta}_{i+1} d_{i+1}}{2(d_i + d_{i+1})} \right), \qquad 1 \le i \le n. \quad (3)$$

where $d_i = M[P](P_{i-1}, P_i)$ is the Nielson metric [4,6], $\hat{\theta}_i = \min(\theta_i, \pi/2)$, and the θ_i is the angle between L_{i-1} and L_i.

Results of Foley's method are sometimes better than those of the centripetal method [3]. Although Foley's method has been claimed to cope with abrupt data distribution well, it also results in similar wiggles to the chord length method

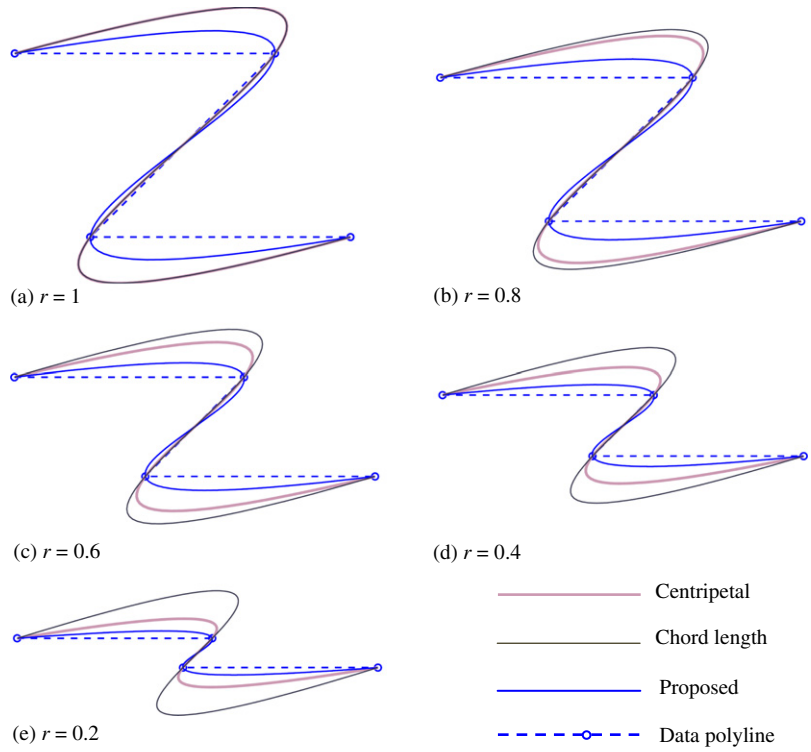


Fig. 2. The interpolating results of different r.

as shown in Fig. 1(d). Other early research on parameterization presented other methods which have fallen out of favor [3]. In addition, the Nielson's metric in Eq. (3) can be simplified using the centripetal or chord length metric [3,4], therefore, we can regard Eq. (3) as a scaling formula on the centripetal or chord length metric. In 1999, Lim proposed a universal method for assigning the parameter value at the maximum of each B-spline basis function to the associated data point [7]. Then he discussed the properties of continuity of this parameterization in 2002 [8]. This method introduces the feature of basis function to replace the geometric features of data. The interpolating results are more natural than those of the uniform, centripetal and Foley methods. The universal method also works well in any degree. However, similar to the drawback of the uniform method, cusps or loops are easily developed in the resulting curves as shown in Fig. 1(f). In addition, it can be observed from Fig. 1(f) that the universal method cannot prove the shape features of the data, such as symmetry and corner shapes. Following Lim, some more recent research has continued to develop the universal method [9,10]. The only example in [9,10] is involved in ship hull surface design. Park [11], however, has shown that the nodes and knot placement of a closed B-spline curve interpolation are shifted to a sequence of a closed data polygon. In his study, the nodes and knot are still initially constructed through one of the uniform, chord length and centripetal methods, then the specific distance is shifted at each node and knot point to obtain a new parameterization. Park focused on solving both even and odd degree closed B-spline curve interpolation problems to a closed data polygon, so a cross-shaped polygon is considered. The works in [9–11] emphasize the solution of a linear system on some specific cases (situations). However, they did not test whether these methods present the geometry of the data polyline, such as symmetry of the distribution. Although the centripetal method seems to be the best compromise [3], it fails to present the overall features of the data distribution.

On the other hand, reproduce or refine the given curves also have been preferred. One of the methods involved in this work is the parameterization optimization to reach the re-parameterization [12–16]. Initial parameterization was needed in these works. Again, the uniform, centripetal, chord, universal or Foley methods were commonly involved. The only different initial parameterization was proposed by Ma and Kruth [14]. They projected the data points to the base surface to compute the data point parameter values. However, the base surface could take any form that should satisfy the unique local mapping, smoothness and closeness properties. Thus data distribution detection was needed before choosing a base surface. For example, if the base curves are chosen as BC1 or BC2, shown in Fig. 1(a), without the feature detection, then some of the parameter values of the data points will be repeated. Another kind of method which re-parameterizes the known curves is proposed by Zhang et al. [17]. They transfer the initial node value to new of each data point, according some geometric relationship of four consecutive data points.

The work we concern here is about the data with unknown nodes, such as the structured free-form data, we should *priori* estimate the node values of the data distribution. Therefore we still focus on selecting the node values of the original data distribution.

To sum up, the centripetal and the chord length methods are preferred among those methods mentioned above, then the uniform method in CAGD applications, respectively. Further, the best choice is the centripetal method with regard to computational effort [3,4]. Even so, the centripetal method still is insufficient for all designs because the interpolating curve will still wander away within the abrupt data distribution as shown in Fig. 2, where the ratio r of the intermediate side length to other sides decreases from 1 to 0.2. This behavior is similar to that of the chord length method.

In this study, we propose a parameterization method which is more suitable for any data, than those known methods. We introduce a briefly-refined method which obtains a more suitable shape for the B-spline interpolation of a free form grid data, especially with regard to abrupt distribution. In our refinement formula, an "additional dragged spacing", which is caused by arc length of the osculating circle at the data point, is added to the

centripetal parameter spacing of each data side to get the new parameter value. The illustration of the refined method will be presented in Section 2.

Aside from the problem of node choosing, the knot placement indeed plays an obviously key role,so that numerous studies have been devoted to this field [18–23]. But in order to further simplify our comparison among these parameterization methods, we adopt the averaging method [1] to set a clamped knot vector as

$$u_0 = \cdots = u_p = \bar{u}_0 \qquad u_{m-p} = \cdots = u_m = \bar{u}_n$$
$$u_{j+p} = \frac{1}{p}\sum_{i=j}^{j+p-1} \bar{u}_i \quad j = 1, \ldots, p. \tag{4}$$

This determination is the predominantly used method in most applications. It also can avoid the loss of generalities and give a nonsingular stable linear system matrix.

Once the interpolating curves are built, investigating the performance of these results is important. A wiggle evaluation is proposed to compare the results of the proposed, uniform, centripetal, chord length and Foley's methods.

The proposed method not only can retain the fine behavior of the centripetal method, but also can significantly promote its results when we interpolate an abruptly-changing data series. Furthermore, the resulting curves of the proposed method remain stable under changing degrees of interpolation. The proposed method is also applied well in tensor-product B-spline surface interpolations.

2. The proposed method

2.1. The refined formula

It is known that, both the interpolating curves of the centripetal and chord length methods oscillate more significantly while the data positions are more abrupt. Although the centripetal method yields obviously less oscillation than the chord length method, we believe that, there still could be an effort made to reduce the oscillations caused by the centripetal method.

The centripetal method is derived based on the physical heuristics of driving a car, the parameter value of each data point as the arrival time of a particle travel event on a curve, in which the spacing of each two adjacent points is proportional to Δ_i. But Δ_i is computed only by the square root of each side norm, disregarding the angles formed by the data distribution [6]. We retrieve the lack of this geometric feature of these angles by introducing a dragging tolerance, where the traveling time spacing is considered to be "dragged" when passing through P_i. Thus we redefine the parameterization formula as

$$\Delta_i = \|L_i\|^{1/2} + e_i \tag{5}$$

by adding e_i as the dragged tolerance, which is computed from the osculating circle at P_i. Since the first term in Eq. (5) is computed by the side norm, it is natural to compute e_i using the other geometric feature, the angle change at P_i.

At first, the discrete osculating circle at P_i needs to be determined. We refer to some works about calculation and applications of the discrete curvature [23–25]. In these works, the discrete curvature is often used to filter the contour from dense and noisy pixels. The circumcircle of three adjacent data points is the most-used conception in methods to compute the discrete curvature of each data point. We adopt this conception to compute the discrete osculating circle.

Recall the tangent T of a curve C is

$$T = \frac{dC}{ds}$$

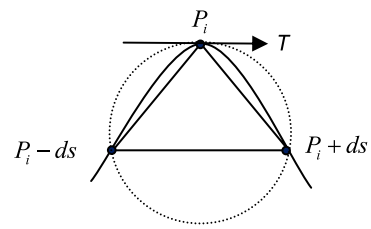


Fig. 3. The approximate tangent T at P_i.

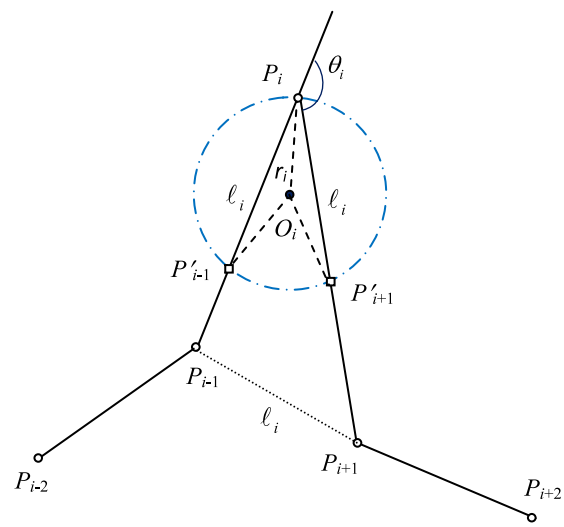


Fig. 4. The circumcircle to compute discrete osculating circle at P_i.

where s is the arc length parameter of the curve. From the view of the points of approximation, Fig. 3 shows that, when ds is small, $P_i - ds$, P_i and $P_i + ds$ can be approximate an isosceles triangle. Thus T at P_i is also the tangent at P_i on the circumcircle of this isosceles triangle. Hence, the circumcircle of this isosceles triangle can be taken as the discrete osculating circle at P_i.

According to this fine approximation, we compute the circumcircle of an isosceles triangle at P_i to get the discrete osculating circle at P_i. First we determine an isosceles triangle at P_i. Take Fig. 4 for an example, recall that $\Delta P_{i-1}P_iP_{i+1}$ is the intuitive choice to compute the discrete osculating circle at P_i. Therefore, we set an isosceles triangle at P_i on $\Delta P_{i-1}P_iP_{i+1}$. We find the shortest side of$\Delta P_{i-1}P_iP_{i+1}$, which is $\overline{P_{i-1}P_{i+1}}$ in this example, and denote this shortest length as ℓ_i. Then set an isosceles triangle $\Delta P'_{i-1}P_iP'_{i+1}$ by setting both $\overline{P'_{i-1}P_i}$ and $\overline{P_iP'_{i+1}}$ equal to ℓ_i. Next we compute the circumcircle O_i of $\Delta P'_{i-1}P_iP'_{i+1}$. On $\Delta P'_{i-1}O_iP_i$, which is divided from $\Delta P'_{i-1}P_iP'_{i+1}$ by the circumradius r_i, $\angle P'_{i-1}O_iP_i = \theta_i$ and $\angle P'_{i-1}P_iO_i = \angle P_iP'_{i-1}O_i = \frac{\pi-\theta_i}{2}$. By the Law of Sine,

$$\frac{r_i}{\sin\left(\frac{\pi-\theta_i}{2}\right)} = \frac{l_i}{\sin\theta_i}. \tag{6}$$

Then by the formulas of trigonometric function, we substitute Eq. (6) as

$$\frac{r_i}{\cos\frac{\theta_i}{2}} = \frac{l_i}{2\sin\frac{\theta_i}{2}\cos\frac{\theta_i}{2}}.$$

Thus we get r_i by

$$r_i = \frac{\ell_i}{2\sin\frac{\theta_i}{2}}. \tag{7}$$

Table 1
Wiggles between the interpolating curve and the data in Fig. 1.

Wiggle	Centripetal	Chord length	Foley	Uniform	Universal	Proposed
W_C	0.850222	3.289140	1.396480	50.437600	167.364000	0.866599

Table 2
Wiggles of the results in Fig. 2.

r	1	0.8	0.6	0.4	0.2
Centripetal	14.56760	13.62680	12.50430	11.09420	9.13731
Chord	14.53030	15.57710	16.88910	18.54980	21.57430
Proposed	9.12355	8.09511	7.11379	6.31456	5.56490

After computing the circumcircle of $\Delta P'_{i-1}P_iP'_{i+1}$, the appropriate osculating circle at P_i is determined by scaling $\ell_i$in order to approximate ds. Assume that

$$ds = \alpha \ell_i. \tag{8}$$

The scale α can be either a variable or a constant. We set α as a constant, thus Eq. (5) forms a linear equation and remains the original properties of centripetal method.

Since ℓ_i is scaled by α to approximate ds, the corresponding osculating radius also be derived to be αr_i. Before arriving at P_i, the additional dragged tolerance is added to the travel time spacing from P_{i-1} to P_i along the curve. The minor arc of the osculating circle of P_i from P'_{i-1} to P_i is equal to $\alpha r_i \theta_i$.

Analogously, the osculating circle at p_{i-1} also affects the time spacing Δ_i in similar way. As a consequence, the dragged tolerance e_i can be written as

$$\begin{aligned} e_i &= \alpha r_i \theta_i + \alpha r_{i-1}\theta_{i-1} \\ &= \alpha \left(\frac{\theta_i \ell_i}{2 \sin \frac{\theta_i}{2}} + \frac{\theta_{i-1}\ell_{i-1}}{2 \sin \frac{\theta_{i-1}}{2}} \right), \quad i = 1, \ldots, n \end{aligned} \tag{9}$$

where α is determined as 0.1. We leave the detail discussion about evaluating α in the later section.

At last, we get the final proposed equation by substituting Eq. (9) for e_i:

$$\Delta_i = \|L_i\|^{1/2} + 0.1 \left(\frac{\theta_i \ell_i}{2 \sin \frac{\theta_i}{2}} + \frac{\theta_{i-1}\ell_{i-1}}{2 \sin \frac{\theta_{i-1}}{2}} \right). \tag{10}$$

By using Eq. (10), the proposed method, the wiggles of the centripetal method in the interpolating results will be decreased. In more advanced cases, the oscillations of the curves by the proposed method are less affected under the interpolating degree change. We will discuss these experiments in the Section 3.

2.2. Interpolation fitness criterion

After building a B-spline, the undesirable outcomes curve interpolation, such as wanders, loops or peaks, indicate that the curve deviates from the data polyline. To filter these poor outcomes, the wiggle analysis between the curve and the given point polyline is considered as an appropriate criterion to judge the performance of these methods. Park [20] uses Hausdorff distance to verify the fitness of curve refinement. Zhang et al. [17] use quadratic precision as a criterion to rebuild a curve. The most significant and intuitive wiggle validation is the distance deviation between the curve and the data polyline. In this paper, the distance from the curve segment to the segment side L_i is simply used to validate the level of wiggles. More the curve wiggles from the side, the larger is the deviation. Therefore, the distance metric

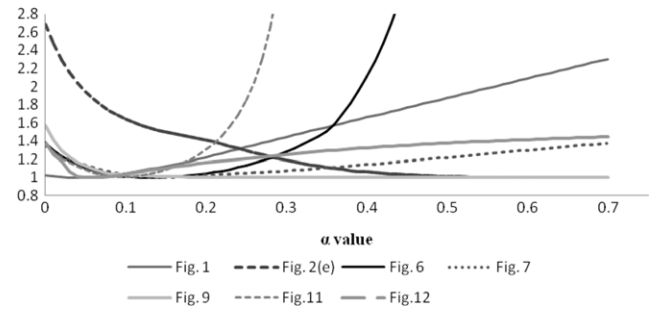


Fig. 5. The distributions of wiggle ratio to optimum of α in different curves.

of deviation between the interpolating curve and given data is defined as

$$D_i = \max_{\bar{u}_i < u < \bar{u}_{i+1}} dist(C(t), L_i), \quad 0 \le i \le n-1 \tag{11}$$

where $dist(P, L)$ is a metric function which sets the value to the minimum distance from a point P to the straight line segment of a vector L.

However, loops or peaks do not be verified, only using Eq. (11). Due to the curvature suddenly raises at the loops or peaks and there are at least two local extreme curvatures if there are wiggles on the arc. We multiple the curvature as a weighting, or also seen as a penalty, to each deviation D_i to present the level of wiggling. Thus the wiggle validation on each side is defined as

$$W_i = (1 + K_i)D_i, \quad 0 \le i \le n-1 \tag{12}$$

where K_i is the maximum of the absolute curvature of the curve arc on this interval. We validate the performance of a curve by averaging W_i as Eq. (13).

$$W_C = \frac{1}{n} \sum_{i=0}^{n-1} W_i. \tag{13}$$

A rising W_C indicates that some arc segments of the curve are further away from the data polyline, or has extraordinary changes of curvatures in this arc. For example, most resulting curves in Fig. 1 are close to the data polyline. But observing Table 1, W_C of the uniform and universal methods are extraordinarily larger than other methods. These results fed back by their performance of loops. In addition, W_C of the chord length is also larger because of its wiggles on some sides of data polyline. However, W_C of the proposed method is also higher than the centripetal method by its high curvatures on some sides. We will discuss these results in detail later. Table 2 shows the wiggles of each case in Fig. 2. As r decreases, more obviously the proposed method outperforms the centripetal and chord methods.

2.3. Analysis of α evaluation

In the Eq. (10), the proposed method, value of α affects the resulting curve critically. We evaluate the value of α from experiments. We optimize α of most curve cases in our study as listed in Table 3. All the tests are interpolated under the 3rd degree. The average of the optimum α is 0.198254. If we ignore the results of

Table 3
The optimum of α value in different cases.

α	Fig. 1	Fig. 2(e)	Fig. 6	Fig. 7	Fig. 9	Fig. 11	Fig. 12	Average
Optimum	0.035997	0.612479	0.127313	0.143271	0.321190	0.095447	0.052078	0.198254

Table 4
Comparisons of wiggles of different α in different cases.

W_c	Fig. 1	Fig. 2(e)	Fig. 6	Fig. 7	Fig. 9	Fig. 11	Fig. 12	Average	W_c ratio to optimum
Optimum α	0.830409	3.398710	4.322260	2.280800	8.484010	3.124730	14.427500	5.266917	1.000000
$\alpha = 0$	0.850222	9.137310	5.952630	3.045410	13.317000	4.270800	19.939900	8.073325	1.532837
$\alpha = 0.09$	0.852881	5.710380	4.383560	2.392420	8.765390	3.140180	14.767200	5.716002	1.085265
$\alpha = 0.1$	0.866599	5.564900	4.350830	2.359310	8.715450	3.130370	14.963100	5.707223	1.083598
$\alpha = 0.2$	1.014780	4.794060	4.505980	2.336370	8.515480	4.072270	16.723800	5.994677	1.138176

Table 5
Wiggles of different α in Fig. 6.

	$\alpha = 0.0$	$\alpha = 0.1$	$\alpha = 0.2$	$\alpha = 0.3$	$\alpha = 0.4$	$\alpha = 0.5$	$\alpha = 0.6$
W_C	5.952630	4.350830	4.505980	5.528430	9.103290	22.329500	61.597400

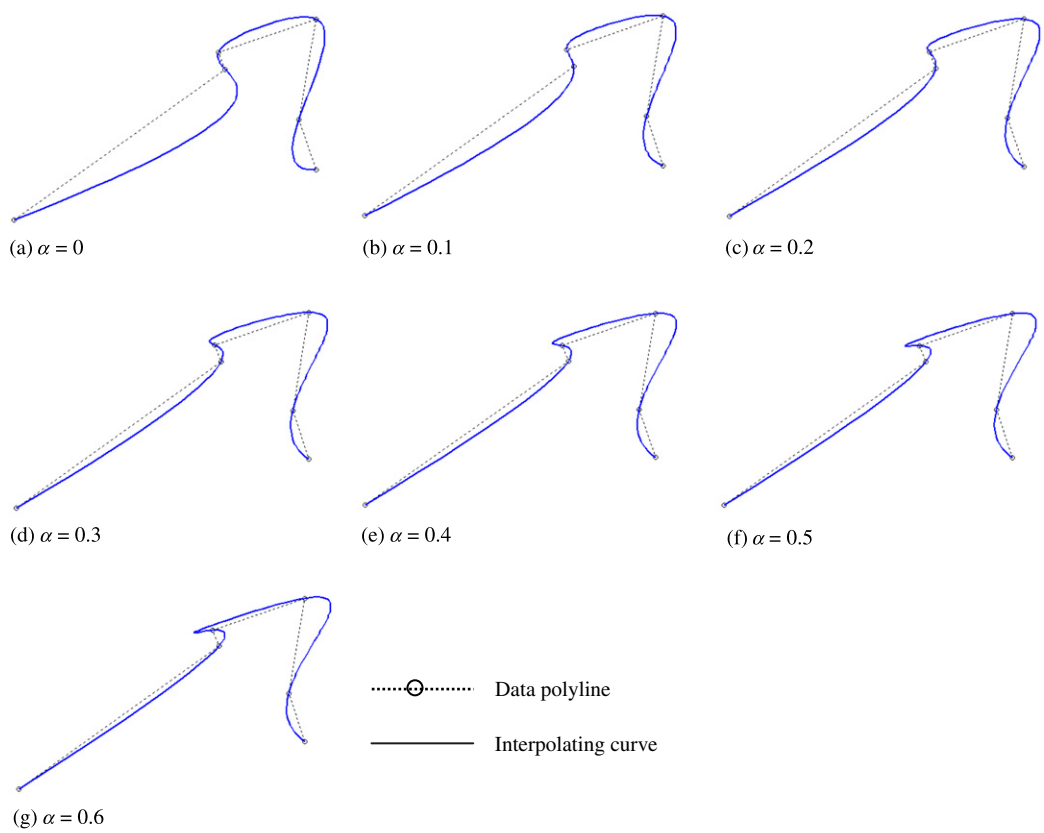


Fig. 6. The interpolating results of different value of α.

Figs. 2(e) and 9, the average of the optimum α is 0.09082. However, compare the wiggles of $\alpha = 0.9$, $\alpha = 0.1$, $\alpha = 0.2$ with the optimum α shown in Table 4, the relative ratio of wiggles of $\alpha = 0.1$ to the optimum indicates that $\alpha = 0.1$ is closest to the optimum. On the other hand, the corresponding distributions of wiggle ratio to optimum in each curve case are demonstrated in Fig. 5. The distributions in Fig. 5 also present that the optimum α is near 0.1 or the wiggle of $\alpha = 0.1$ is very close to the optimum. So we conclude that $\alpha = 0.1$ is a reasonable value in our proposed method. One of our experiments is shown as Fig. 6 and the corresponding wiggle results are listed in Table 5.

3. Curve implementations

The performance of the proposed method on curve experiments is discussed in this section. The interpolation results of the proposed method are compared with those obtained by the centripetal, chord length, Foley's, uniform and universal methods.

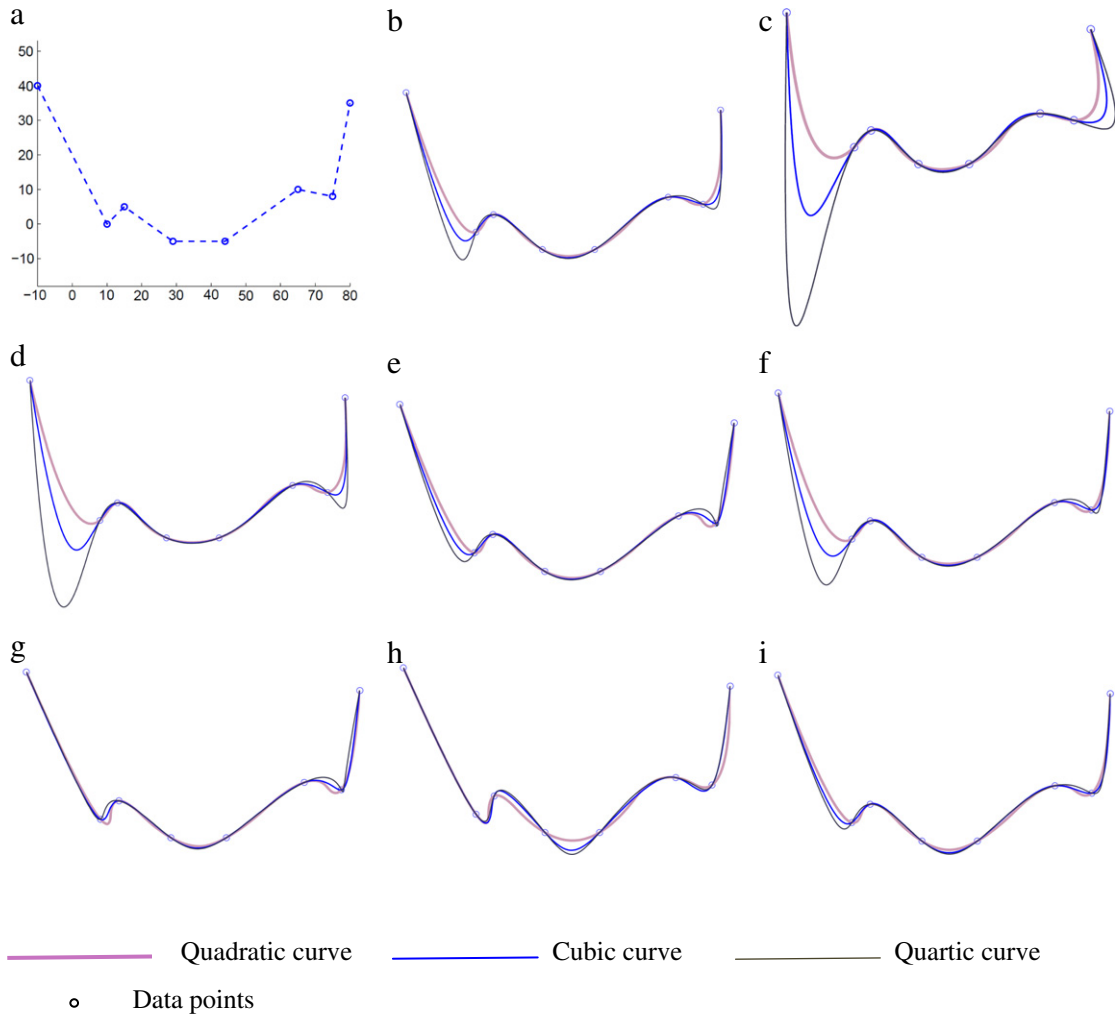


Fig. 7. Interpolation of different degrees using different methods: (a) data polyline (b) centripetal, (c) chord length, (d) Foley, (e) F-CEN, (f) F-CHD, (g) uniform, (h) universal and (i) proposed.

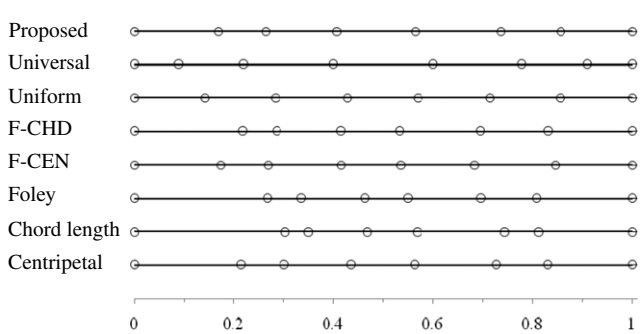


Fig. 8. Parameter diagram of each method in Fig. 7.

In more detail, Foley's method is subdivided into three methods by using Nielson (Foley), centripetal (F-CEN) and chord length (F-CHD) metric. In planar curves, when interpolation is applied to the given points, less deviation always implies fewer wiggles, and even fewer loops. Hence the wiggle evaluation between the resulting curve and the data points is one of our concerns. Furthermore, the influence on wiggles by degree of interpolation is also included in this discussion. We discuss the conditions under degrees 2–4, which are most applied in CAGD field.

3.1. Abrupt-direction changing sides in data

It is known that the chord length method makes wiggles easily in the segments of the abrupt-direction changing data. In other words, the curves will wander away from the polyline when both the turn angles of adjacent data segments are contrary and when their length difference is significantly large. The examples in Fig. 2(a)–(e) are Z-shaped data distribution, which is a simple test for an abrupt distribution. As the ratio r of the intermediate side length to other sides decreases from 1 to 0.2, the points distribute more abruptly. The resulting curves in Fig. 2(a)–(e) show that the chord length method wiggles more sharply while the ratio gets small. From Table 2, the values of W_C show that the centripetal method is less affected by the distribution, but the proposed method is more superior to the centripetal method. Notice that, the results of the centripetal and the chord length are the same of $r = 1$ due to their algorithm, Eq. (2).

Furthermore, the example in Fig. 7 distributes abruptly in two end regions and have small wave in the interior. Both the curve results and the wiggle validation reflect that, the chord length and Foley's method wiggle most in the two sides of the data, and then the centripetal method also wanders a little as the interpolating degree increases. It is also observed from Table 6 that the wiggle values of chord length, Foley, F-CHD, uniform and the universal methods raises suddenly by occurring peaks,

Table 6
Wiggles of the results in Fig. 7.

Degree	Centripetal	Chord length	Foley	F-CEN	F-CHD	Uniform	Universal	Proposed
2	2.147880	3.723380	2.450730	1.794740	2.194170	2.233700	3.348640	2.021620
3	3.045410	8.824730	4.380490	2.408130	3.333400	1.965250	3.996550	2.359310
4	5.145940	27.479800	11.271300	6.014270	6.003610	3.761170	4.035860	3.023750

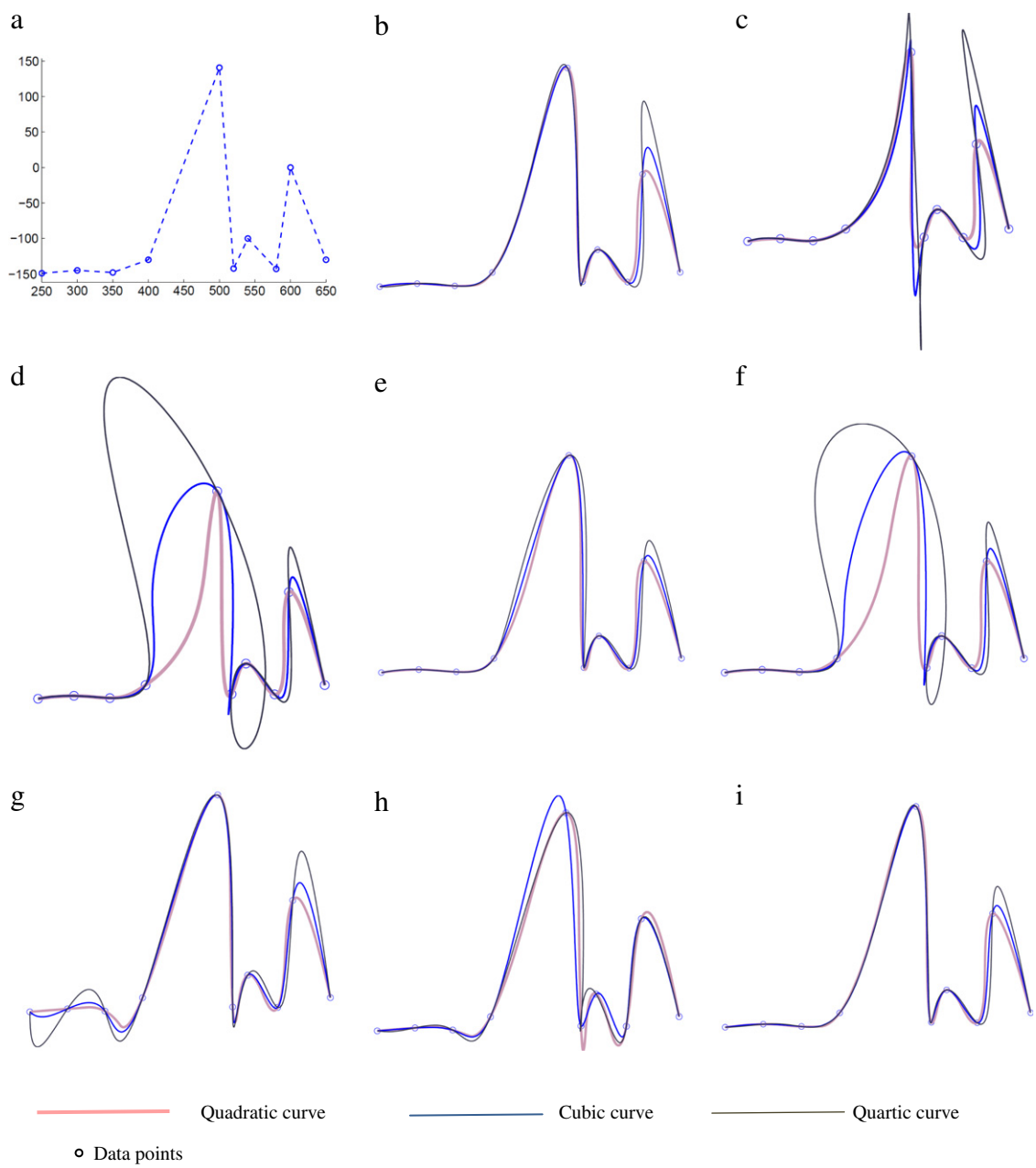


Fig. 9. Interpolation of different degrees using different methods: (a) data polyline (b) centripetal, (c) chord length, (d) Foley, (e) F-CEN, (f) F-CHD, (g) uniform, (h) universal and (i) proposed.

or wandering away. The parameters of data computed by each method are also listed as shown in Fig. 8. This diagram is cross-referred as the performances. It can be seen that, the methods which have the similar performances have the similar parameter distributions. All corresponding tables of the parameter values of Fig. 8 and other examples are attached in Appendix A. In addition, the listed parameters are computed in degree 3, the parameters do not change under different degrees except the universal method. We score 1–8 to the order of the performances of these methods as W_C raising. In this case as shown in Table 7, we see the proposed method get the minimum score 1.7 by averaging the scores of degree 2 to 4, that means it performs best and most stably.

Fig. 9 shows another extreme example in which, the first part of the data is nearly equal-spaced and smooth, and the second part distributes abruptly. This example indicates that the proposed method outperforms others. It deserves to mention that, the wiggle

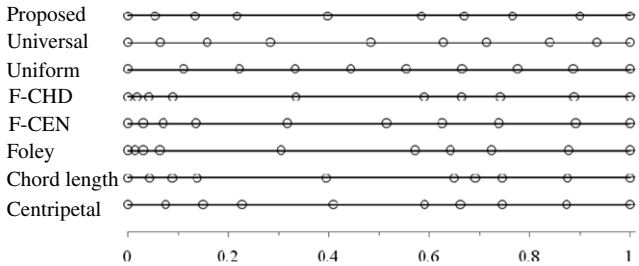


Fig. 10. Parameter diagram of each method in Fig. 9.

is suddenly high of Foley's method in degree 3 and the chord length in degree 4 due to their peaks, even stays as cusps. From the parameter diagram in Fig. 10, we observe that the extra non-uniform distribution of the parameters, such as the F-CHD and Foley methods, may have lead to the unexpected wiggles in results. The wiggles of the results is listed as Table 8. In this case, the average performance score of the proposed method is 1.0. This perfect score indicates the proposed method outperforms other methods in the case of Fig. 9. The average performance scores of all methods of all our curve cases are listed in Table 11.

3.2. Near-equal spacing data

In the case where the data is almost smooth, as shown in Fig. 11, the proposed method behaves well under each degree. F-CHD and the proposed methods perform well. The proposed method still gets score of 1.3, which is a little better than the score of 1.7 of F-CHD, as shown in Table 11. Notably, wanders of the centripetal method in the front segments make its higher wiggle value. Obviously, high wiggle values of Foley's method reflects the loop or deviation. High wiggle values of the uniform and the universal methods are corresponding to their peaks, even cusps. Generally, the outcomes of most of these methods are accepted, we leave the parameter diagram out.

Another test in Fig. 12(a) is a mixed one, in which one part of the data has long chords but is less abrupt, and one of the parts is where

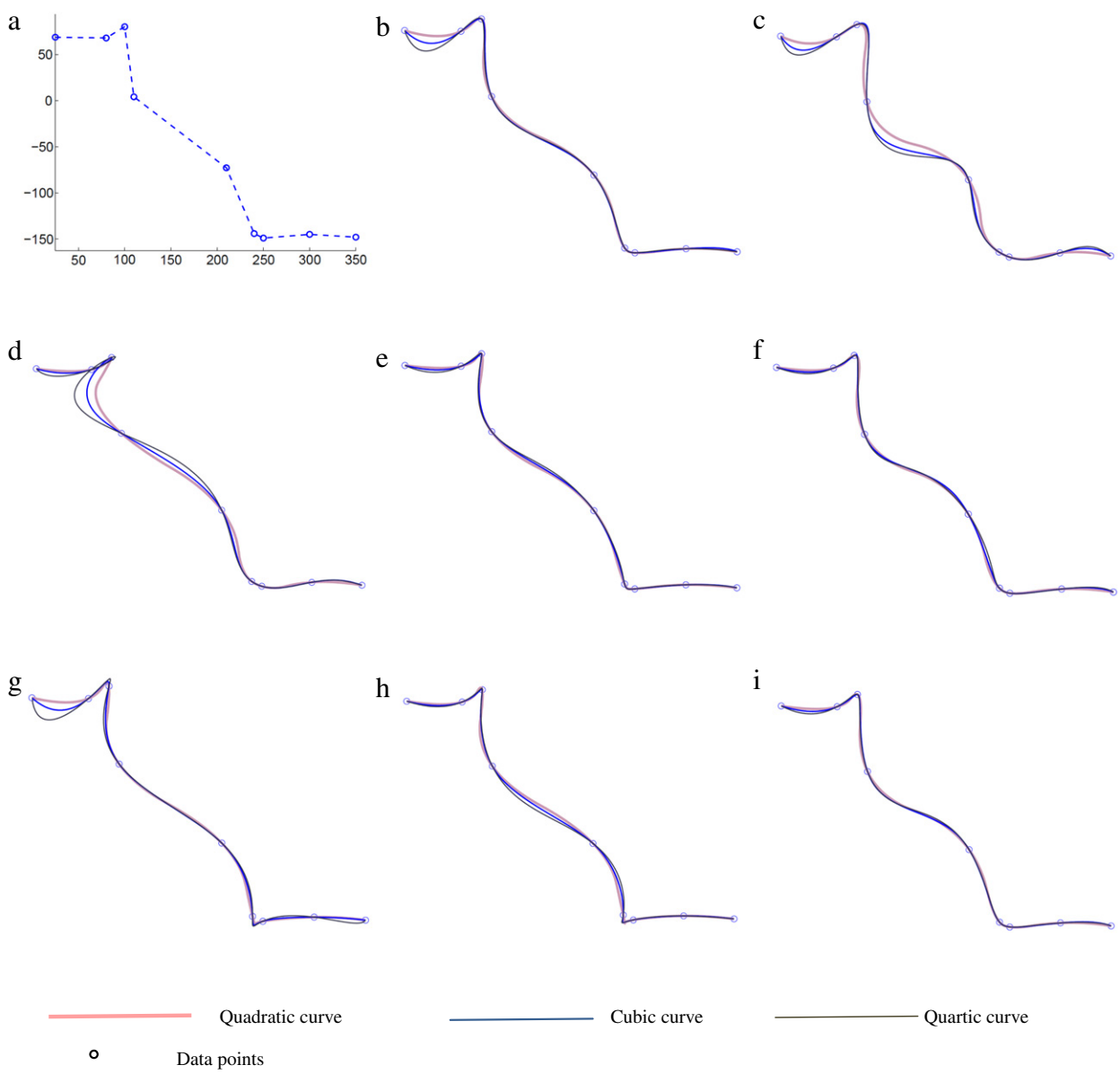


Fig. 11. Interpolation of different degrees using different methods: (a) data polyline (b) centripetal, (c) chord length, (d) Foley, (e) F-CEN, (f) F-CHD, (g) uniform, (h) universal and (i) proposed.

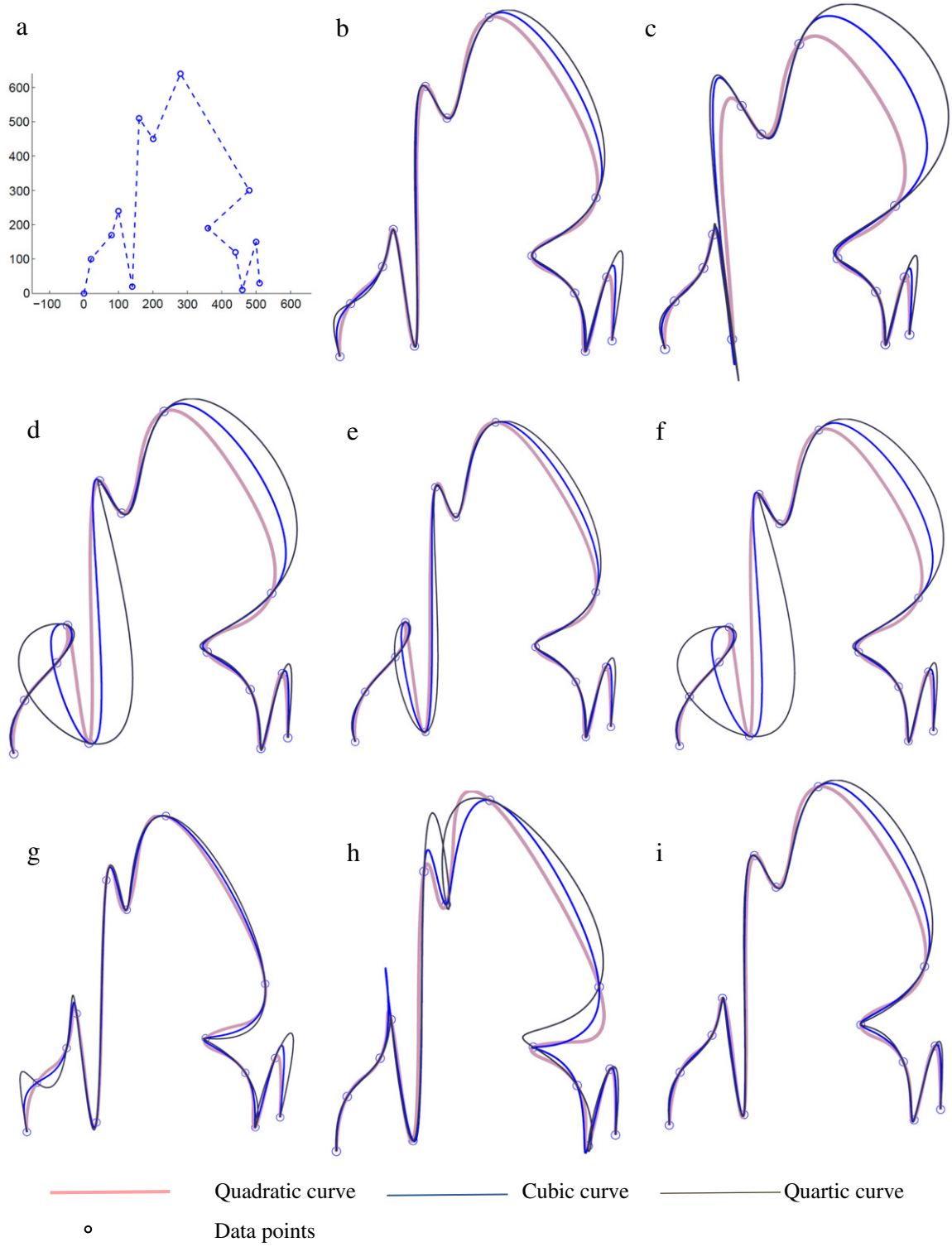


Fig. 12. Different degree of interpolation by different methods: (a) the data polyline (b) centripetal, (c) chord length, (d) Foley, (e) F-CEN, (f) F-CHD, (g) uniform, (h) universal and (i) proposed.

the successive data side vectors turn in the same direction, or say U-shape. In this case, the results of the centripetal method are close to the proposed method. However, the centripetal method wiggles much in degree 4. As a result, the proposed method is stable and wiggles least under degree changing in contrary. Foley, F-CEN and F-CHD have higher wiggles due to their loops. The uniform and universal methods still have peaks to make much higher wiggle values. The results of chord length method are affected by either wanders or peaks. Wiggle values in Table 10 show these results significantly. In this case, the average score of the proposed method is also the best. The diagram in Fig. 13 still shows that, the big wiggles of Foley, F-CEN and F-CHD accompany their extra non-uniform parameter distributions. These results indicate that Foley's formula does not yet fit all the kinds of sharp data distribution yet.

From the scores shown in Table 11, we can conclude that the proposed method is best than other methods in average of all curve cases in this study. However, in Fig. 1 of a smooth example

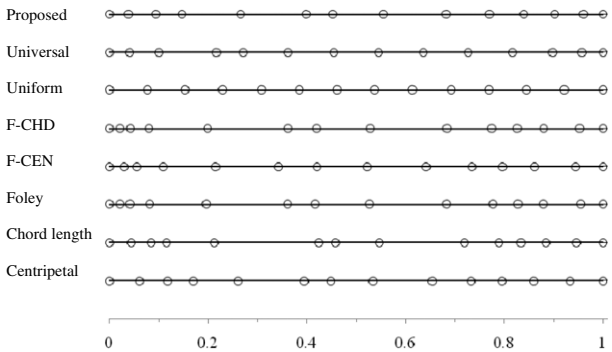


Fig. 13. Parameter diagram of each method in Fig. 12.

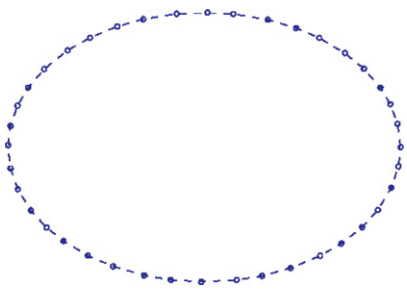


Fig. 14. Ellipse and $n = 40$ points.

addressed by Lee [2], the centripetal method works best, then the proposed method. This outcome indicates that when the data tends to "U" or "rounded" shape, the proposed method may have less effect to interpolation than the centripetal method. Even though, the curve performance of the proposed method still nears to that of the centripetal method.

Additionally, there are little unexpected oscillations of Foley's method in the regions between P_5–P_6 and P_{10}–P_{11} in Fig. 1, the result of which is similar to the chord length method. These results also indicate that Foley's method may behave similarly to the chord length method when going through longer chords. Other examples in Figs. 7 and 9 also show this property of Foley's method. Fig. 1(e) and (f) show the loops occurring in the resulting curves of the uniform and universal methods. Furthermore, Appendix B lists the

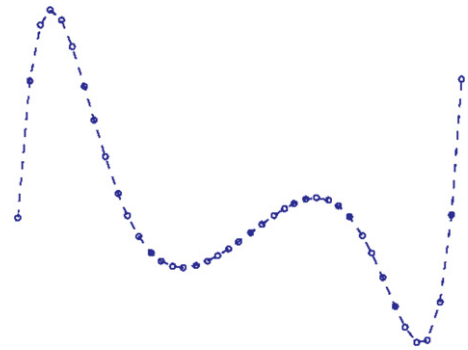


Fig. 16. Polynomial curve and $n = 40$ data points.

wiggle tables of Figs. 7, 9, 11 and 12 with degree = 3 as examples of details of wiggle distribution of different methods to each curve case.

3.3. The influence of degree increases

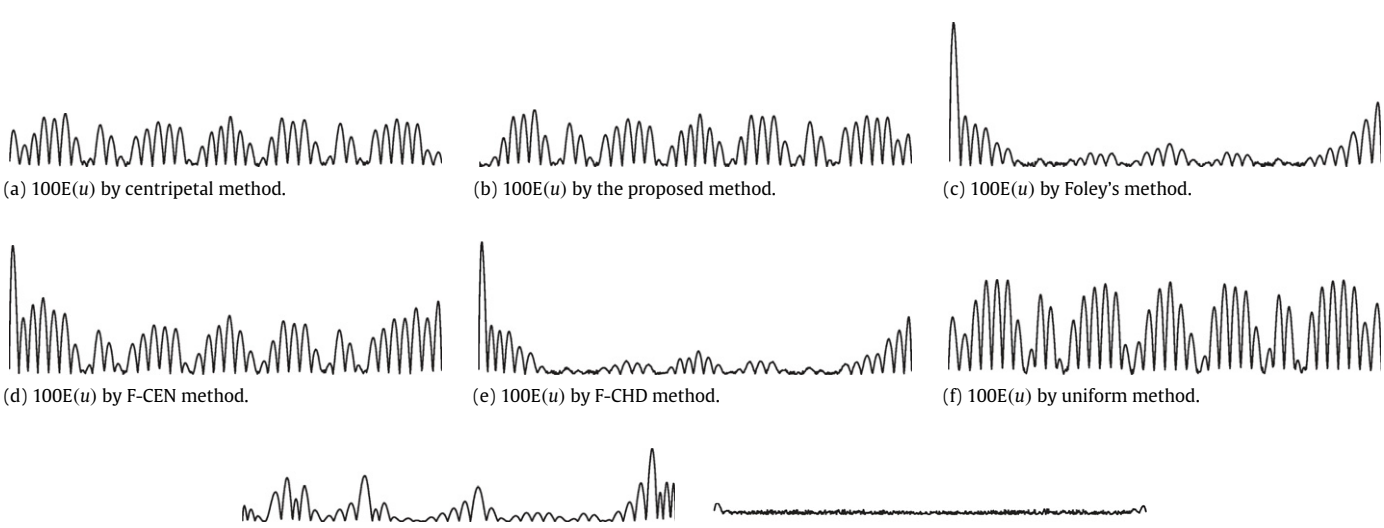
In general, as the degree of interpolation increases, the resulting curve wanders more. This is a common feature of spline curves, regardless of the parameterization method used. From Tables 6, 8 to 10, we can see that, there occur a suddenly higher W_c of each method under degree changing, except the proposed method. This shows that the proposed method is more stable among other methods under degree changing.

As a brief summary, the proposed method not only certainly reduces the oscillations of the centripetal method on the abrupt-changing data, but it also maintains the good properties of the centripetal method, such as few loops, intersections, and geometric features of symmetry and corner keeping.

3.4. Interpolation error comparison

In addition, another method to measure the fairness of the curve fitting is the interpolation error. Here we try two examples to justify the effect of the proposed method. All the interpolating curves are cubic. In the first example, the data points used in the comparison are taken from the ellipse, $F(t) = (x(t), y(t))$ defined by

$$\begin{aligned} x(t) &= 3\cos(2\pi t) \\ y(t) &= 2\sin(2\pi t) \end{aligned} \tag{14}$$



(a) 100E(u) by centripetal method.
(b) 100E(u) by the proposed method.
(c) 100E(u) by Foley's method.
(d) 100E(u) by F-CEN method.
(e) 100E(u) by F-CHD method.
(f) 100E(u) by uniform method.
(g) 10E(u) by universal method.
(h) 100E(u) by chord length method.

Fig. 15. Error curves to ellipse by different methods with $n = 40$.

Table 7
Score table of the results in Fig. 7.

Degree	Centripetal	Chord length	Foley	F-CEN	F-CHD	Uniform	Universal	Proposed
2	3	8	6	1	4	5	7	2
3	4	8	7	3	5	1	6	2
4	4	8	7	6	5	2	3	1
Average	3.7	8.0	6.7	3.3	4.7	2.7	5.3	1.7

Table 8
Wiggle of the results in Fig. 9.

Degree	Centripetal	Chord length	Foley	F-CEN	F-CHD	Uniform	Universal	Proposed
2	7.80045	15.36440	15.31400	7.15588	8.88339	19.92250	37.48740	6.79889
3	13.31700	205.27200	7891.06000	9.82799	295.79100	26.72420	11.67150	8.71545
4	35.61490	2528.70000	56.12690	21.98540	39.49800	53.83530	100.44300	15.74220

Table 9
Wiggle of the results in Fig. 11.

Degree	Centripetal	Chord length	Foley	F-CEN	F-CHD	Uniform	Universal	Proposed
2	3.52484	5.19771	15.40550	4.37097	2.75000	25.31930	221.81700	2.95941
3	4.27080	8.44296	10.65940	7.49956	3.27053	7.84729	8.12620	3.13037
4	5.47828	9.83278	12.10220	4.85518	3.57422	13.14290	8.55764	3.55510

Table 10
Wiggles of the results in Fig. 12.

Degree	Centripetal	Chord length	Foley	F-CEN	F-CHD	Uniform	Universal	Proposed
2	13.91050	58.87360	16.40410	33.62290	18.53220	22.64530	21.56110	12.93650
3	19.93990	448.97700	20.61840	16.50440	21.46260	33.67520	532.97100	14.96310
4	50.57490	1224.30000	50.72960	56.32270	81.81730	50.39960	70.46850	22.02270

Table 11
Score tables.

Average score	Centripetal	Chord length	Foley	F-CEN	F-CHD	Uniform	Universal	Proposed
Fig. 1	1.0	4.0	2.7	–	–	5.0	6.0	2.3
Fig. 7	3.7	8	6.7	3.3	4.7	2.7	5.3	1.7
Fig. 9	3.3	6.7	6.3	2.0	5.0	5.7	6.0	1.0
Fig. 11	3.3	5.7	7.0	3.7	1.7	6.7	6.7	1.3
Fig. 12	2.7	7.7	3.7	4.7	5.3	4.7	6.3	1.0
Average	2.8	6.22	5.42	3.85	3.35	5.16	6.2	1.52

where the parameter $t \in [0, 1]$. Refer to [17], we compute the n data points by defining t_i as follows:

$$t_i = [i + \lambda \sin(i^*(n - i))]/n \quad i = 0, 1, 2, \ldots, n \tag{15}$$

where $\lambda = 0.1$, which makes the data points distributed non-uniformly. Fig. 14 is the ellipse Eq. (14) and the circle symbols which denote the 40 data points.

The eight methods are compared by the interpolation error $E(u)$ [17], defined by

$$E(u) = \min_t\{|P(u) - F(t)|\}$$
$$= \min_t\{|P_i(u) - F(t)|, t_{i-1} \le t < t_i\}, \quad i = 0, 1, 2, \ldots, n-1$$

where $P(u)$ is one of the B-spline curve construct by the eight methods, $P_i(u)$is the corresponding part of $P(u)$ on the subinterval $[t_{i-1}, t_i)$. The value of $\mathrm{E}(t)$ means the closest distance from a point $P(u)$ to the curve $\mathrm{F}(t)$.

All methods are compared on the data points defined by Eq. (15), where $n = 10, 40, 70, 100$. The resulting curves are not plotted, because the shapes are almost the same. The maximum of $E(u)$ and the corresponding wiggle evaluation are listed in Tables 12 and 13, respectively. For $n = 40$, the error curves of each method are shown in Fig. 15.

The second example is to interpolate a polynomial of degree 7. The polynomial is defined by

$$y = 0.1x^7 - 0.3x^5 - 0.5x^3 + 0.2x^2 + 0.7x + 2 \tag{16}$$

where $x \in [-2, 2]$. We set data point $(x_i, y_i) = (x(t_i), y(t_i))$ defined

$$\begin{aligned} x(t_i) &= -4 + 2t_i \\ y(t_i) &= F(x(t_i)) \end{aligned} \quad i = 0, 1, 2, \ldots, n \tag{17}$$

where t_i is also defined as Eq. (15), and $n = 10, 40, 70, 100$, too. The plot of the polynomial is shown in Fig. 16 where the circle symbols still denote the data points.

The global interpolation process without constraints in this study is used to interpolate this polynomial. All methods are still compared on the data points defined by Eq. (15). The maximum of $E(u)$ and the corresponding wiggle evaluation are listed in Tables 14 and 15, respectively. For $n = 40$, the error curves of each method are shown in Fig. 17.

In the ellipse example, the chord length outperforms other methods both in wiggle and interpolation error evaluation. This result can be corresponding to the "rounded" property of the chord length method. The proposed method is finer than the centripetal method in wiggle meanwhile the interpolation error of the proposed method is larger than the centripetal method. These results indicate the proposed method is better than the centripetal method in considering both the deviation and the curvature at the same time. These results also show that, the proposed method has less effects than the centripetal method in "rounded" distributed data. In the second example of polynomial,

Table 12
Maximum absolute errors on ellipse.

Error	Centripetal	Chord length	Foley	F-CEN	F-CHD	Uniform	Universal	Proposed
$n = 10$	0.0355944	0.0235466	0.0579757	0.0534443	0.0309136	0.0864951	0.0734089	0.0545414
$n = 40$	0.0012068	0.0002594	0.0033410	0.0029888	0.0030817	0.0021803	0.0165740	0.0013250
$n = 70$	0.0010038	0.0000597	0.0003491	0.0011518	0.0003502	0.0018094	0.0061667	0.0010666
$n = 100$	0.0005953	0.0000426	0.0002378	0.0007141	0.0002296	0.0011431	0.0052940	0.0006449

Table 13
The wiggles of each method to Ellipse data polygon.

Wiggle	Centripetal	Chord length	Foley	F-CEN	F-CHD	Uniform	Universal	Proposed
$n = 10$	0.184425	0.179928	0.206821	0.187656	0.181445	0.212509	0.185615	0.179063
$n = 40$	0.011241	0.011152	0.011498	0.011526	0.011446	0.011610	0.016662	0.011192
$n = 70$	0.003704	0.003616	0.003626	0.003698	0.003626	0.003993	0.005143	0.003701
$n = 100$	0.001817	0.001763	0.001768	0.001841	0.001768	0.002073	0.567947	0.001816

Table 14
Maximum absolute errors.

Error	Centripetal	Chord length	Foley	F-CEN	F-CHD	Uniform	Universal	Proposed
$n = 10$	0.124885	0.306340	0.535403	0.384762	0.496299	0.119786	0.166788	0.108492
$n = 40$	0.007388	0.174604	0.033382	0.007448	0.004758	0.006257	0.063755	0.007558
$n = 70$	0.004037	0.005508	0.007259	0.001889	0.001996	0.002645	0.018421	0.004021
$n = 100$	0.001462	0.002259	0.000603	0.000946	0.000612	0.001132	0.016092	0.001455

Table 15
Wiggles of each method to a polynomial of degree 7.

Wiggle	Centripetal	Chord length	Foley	F-CEN	F-CHD	Uniform	Universal	Proposed
$n = 10$	0.833555	0.924647	2.363140	2.049170	2.731050	0.768988	0.762752	0.764749
$n = 40$	0.044793	0.064724	0.048560	0.047866	0.044268	0.051650	0.288798	0.044807
$n = 70$	0.012790	0.013597	0.013119	0.012339	0.012452	0.012753	0.057593	0.012786
$n = 100$	0.005912	0.006215	0.006188	0.006018	0.005917	0.006174	4.727960	0.005912

although the centripetal seems to be finer than the proposed method with $n = 40$ in both of wiggle evaluation and interpolation error, the difference between these two methods is very tiny. This tiny difference also can be validated by the error plot in Figs. 15 and 17. The error curves of the centripetal and the proposed method are almost the same in both Figs. 15 and 17. On the other hand, as the number of the data points increases, the shape of the represented curves of those methods are almost the same. When $n = 40$, the resulting curves of the chord length and universal method are obviously different from other methods. The resulting curves of four methods with $n = 40$ are demonstrated in Fig. 18. From Table 15, the much higher wiggle of the universal method with $n = 40$ validates its peak significantly. Meanwhile, from Table 14, the larger error of the chord length indicates the wander of the resulting curve. These two examples also indicate that, the proposed method performs well under the interpolation error comparison. Furthermore, the results of wiggle validation are close to those of the interpolation error.

4. Surface implementations

For a (p, q)th-degree global B-spline surface interpolation on $(n+1) \times (m+1)$ data grid $\{P_{i,j}\}$, $i = 0, \ldots, n$, and $j = 0, \ldots, m$, where $n \geq p$, $m \geq q$ as the form

$$P_{i,j} = S(\bar{u}_i, \bar{v}_j) = \sum_{k=0}^{n} \sum_{l=0}^{m} B_{k,l} N_{k,p}(\bar{u}_i) N_{l,q}(\bar{v}_j) \tag{18}$$

where the $\{B_{k,l}\}$, $k = 0, \ldots, n$, $l = 0, \ldots, m$, are the control points, and $N_{k,p}$ and $N_{l,q}$ are the basis functions. For computing the control points, we adopt Piegl and Tiller's algorithms of global surface interpolation [1]. We use the same parameterization method to determine the values for the $(\bar{u}_i, \bar{v}_j)$ of each $P_{i,j}$. First, choose a fixed parameterization method, compute parameters $\bar{u}_i^j$ of $P_{i,j}$ $(i = 0, \ldots, n)$ for each j, and then obtain each $\bar{u}_i$ by averaging across all $\bar{u}_i^j$, $j = 0, \ldots, m$. that is

$$\bar{u}_i = \frac{1}{m+1} \sum_{l=0}^{m} \bar{u}_i^j \quad i = 0, \ldots, n.$$

Then determine the corresponding knot vector U also by Eq. (4). Each $\bar{v}_l$ and the knot vector V are computed in the analogous way. Then use Piegl and Tiller's algorithms to obtain the control points.

4.1. Wiggle evaluation

Evaluating the wiggles between the interpolating surface and the data grid is more complicated. Herein we extend the concept of the evaluation of Eq. (13) in the plane curve interpolation onto the surface interpolation. Therefore, we need define the deviation of the surface to data grids, and the curvature.

First we define the distant deviation of the interpolating surface to data grids. As the diagram shown in Fig. 19, for a data patch $A_{i,j} = [P_{i,j}, p_{i+1,j}] \times [P_{i,j}, P_{i,j+1}]$ which is determined on parameter patch $R_{i,j} = [\bar{u}_i, \bar{u}_{i+1}] \times [\bar{v}_j, \bar{v}_{j+1}]$, if a surface point $S(u, v)$, where (u, v) belongs to $R_{i,j}$,the distance of $S(u, v)$ to $A_{i,j}$, denoted as $D(S(u, v), A_{i,j})$, is classified into two conditions to compute:

(1) (u, v) is on the boundary of $R_{i,j}$. Suppose that $u = \bar{u}_i$ and $\bar{v}_j < v < \bar{v}_{j+1}$, then (u, v) is on the one boundary of $R_{i,j}$.

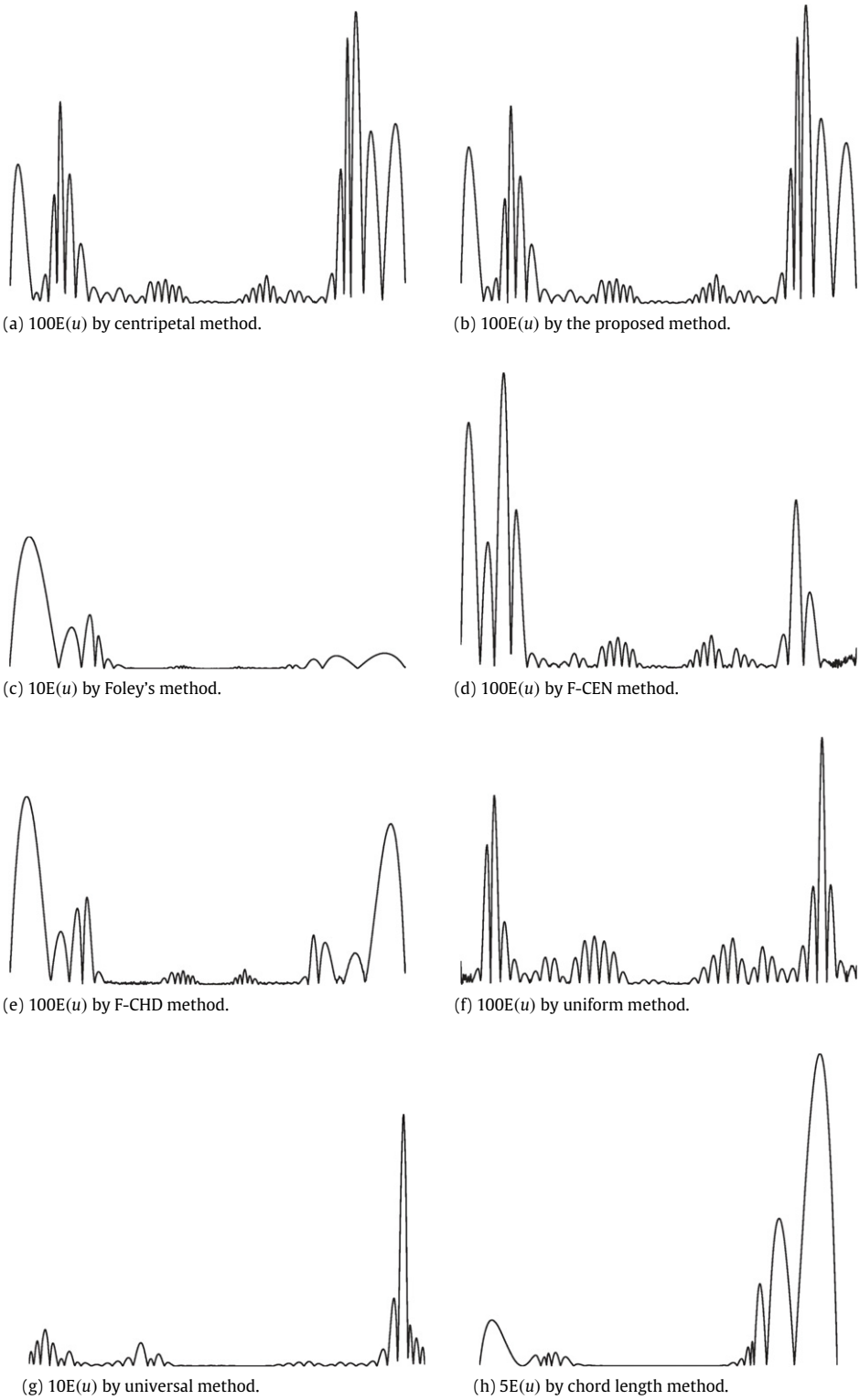
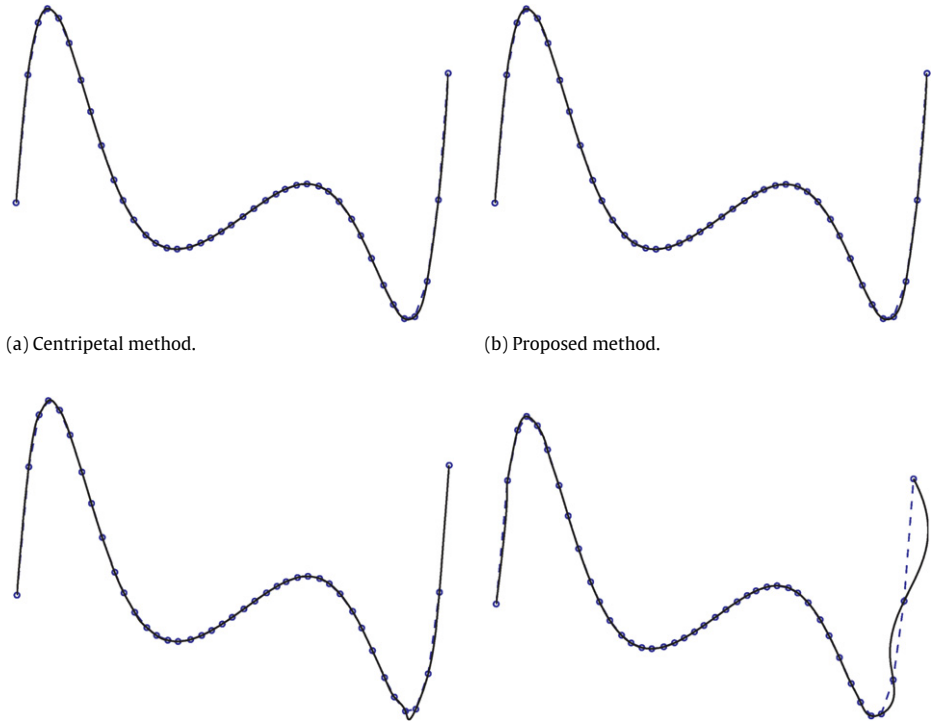


Fig. 17. Error curves to polynomial by different methods with $n = 40$.

The point $S(u, v)$ can be treated as a curve point resulting from an interpolation of the segment $\overline{Q_{i,j}Q_{i,j+1}}$, and thus we set $D(S(u, v), A_{i,j})$ to equal $dist(S(u, v), \overrightarrow{Q_{i,j}Q_{i,j+1}})$, which is as the same metric function used in Eq. (11). The deviation of $S(u, v)$ on the other three boundaries of $A_{i,j}$ is computed in the same way.

(2) (u, v) is in the interior of $R_{i,j}$. As shown the diagram in Fig. 10, $A_{i,j}$ probably is not a plane. Thus, the approximation method is the choice used to compute the deviation. First we approximate a plane $A'_{i,j}$ to $A_{i,j}$ by averaging four planes which are respectively derived by any three of four vertices on $A_{i,j}$. Next, the distance from $S(u, v)$ to $A'_{i,j}$ is assigned as the deviation $D(S(u, v), A_{i,j})$ of $S(u, v)$. The farther the



(a) Centripetal method. (b) Proposed method.

(c) Universal method. (d) Chord length method.

Fig. 18. The resulting curves of four methods with $n = 40$.

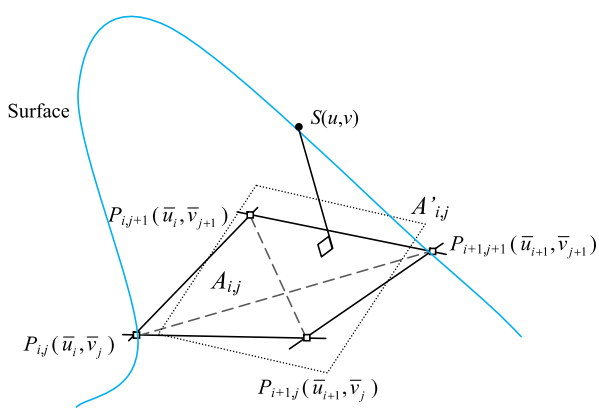


Fig. 19. The diagram of a surface point $S(u, v)$ above a patch $A_{i,j}$.

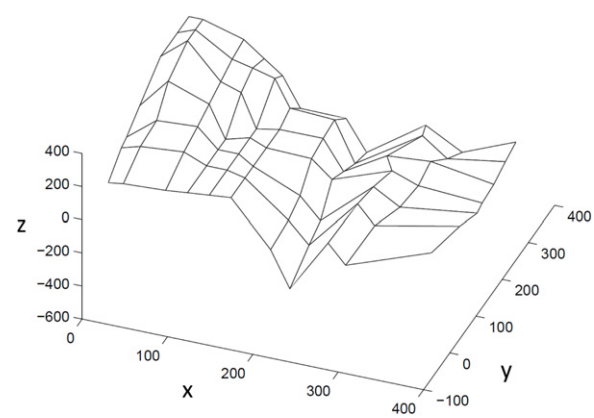


Fig. 20. The test data grids.

$S(u, v)$ is from $A'_{i,j}$, the larger the $D(S(u, v), A_{i,j})$ increases. In addition, such definition guarantees that the orthogonal distance from $S(u, v)$ to $A_{i,j}$ if $A_{i,j}$ is exact a plane. Analogous to the determination in Eq. (11), the deviation of an interpolating surface to the data grids is defined as (Fig. 19):

$$D_{i,j} = \max_{\bar{u}_i \le u \le \bar{u}_{i+1}, \bar{v}_j \le v \le \bar{v}_{j+1}} D(S(u, v), A_{i,j}),$$
$$\forall 0 \le i \le n-1, 0 \le j \le m-1. \tag{19}$$

Second, we choose Gaussian curvature to be the weighting factor for its geometric meaning on the bending of a surface. Then the wiggle value on each patch is defined as

$$W_{i,j} = (1 + K_{i,j})D_{i,j}, \quad 0 \le i \le n-1,\ 0 \le j \le m-1, \tag{20}$$

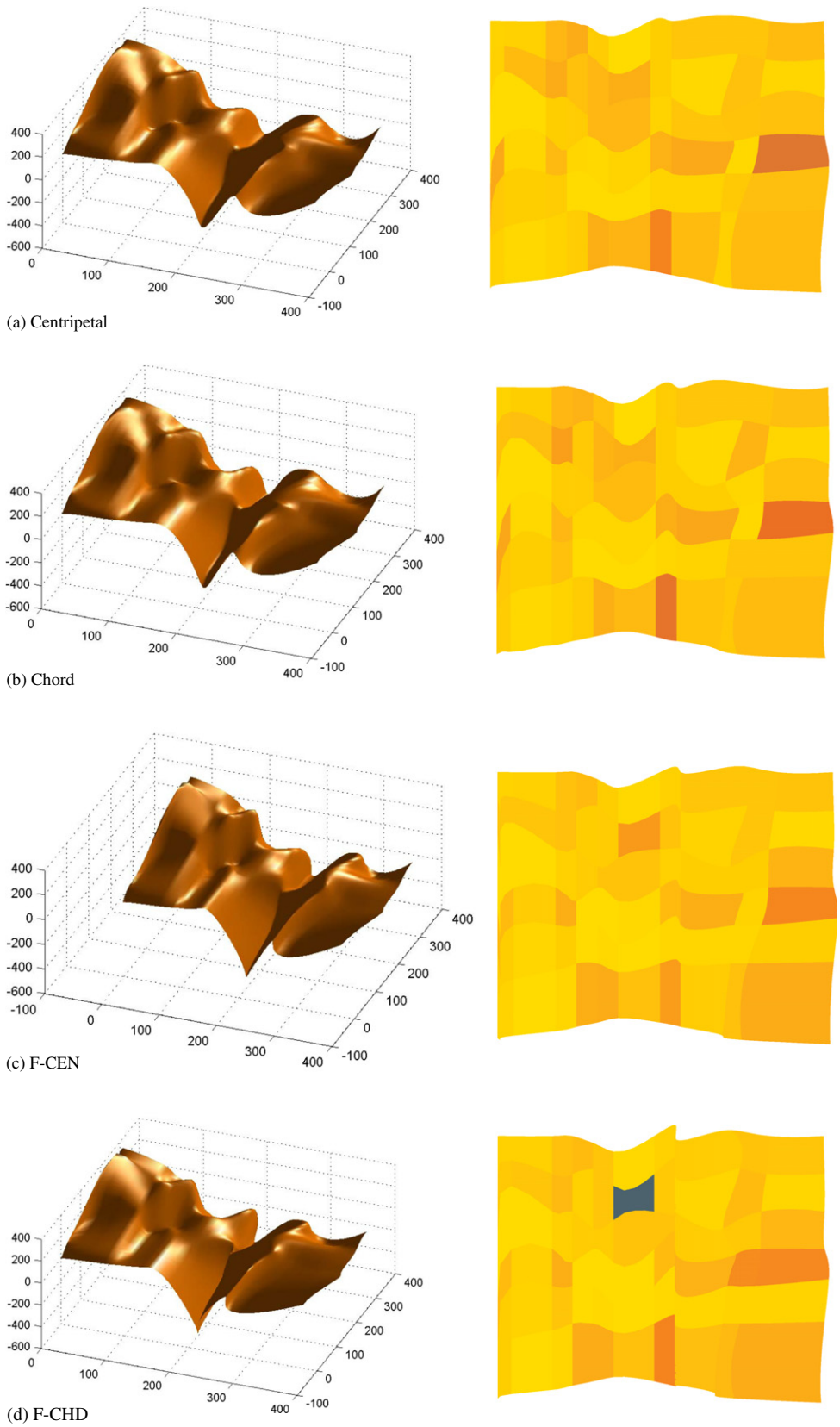
where $K_{i,j}$ is the maximum of the absolute Gaussian curvature on the patch $R_{i,j}$. Analogously, using Eq. (20), the wiggle value of a surface is determined as

$$W_s = \frac{1}{mn} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} W_{i,j}. \tag{21}$$

4.2. Case study

We experiment on a test data and a scanned free-form surface. We compare the results of cubic B-spline interpolating surfaces using those parameterization methods in curve implementations excluding Foley's method with Nielson's metric for its limit to 2D plane curve. All of these interpolating surfaces are presented in two kinds of shapes: one is brushed by one color, and the other is shaded according to the wiggle values of each patch to clearly the wiggle distribution of each method.

The test data shown in Fig. 20 are designed for an abrupt case. The interpolating surfaces and their corresponding wiggle of each



(a) Centripetal

(b) Chord

(c) F-CEN

(d) F-CHD

Fig. 21. Surface interpolation and its contour plot with different methods.

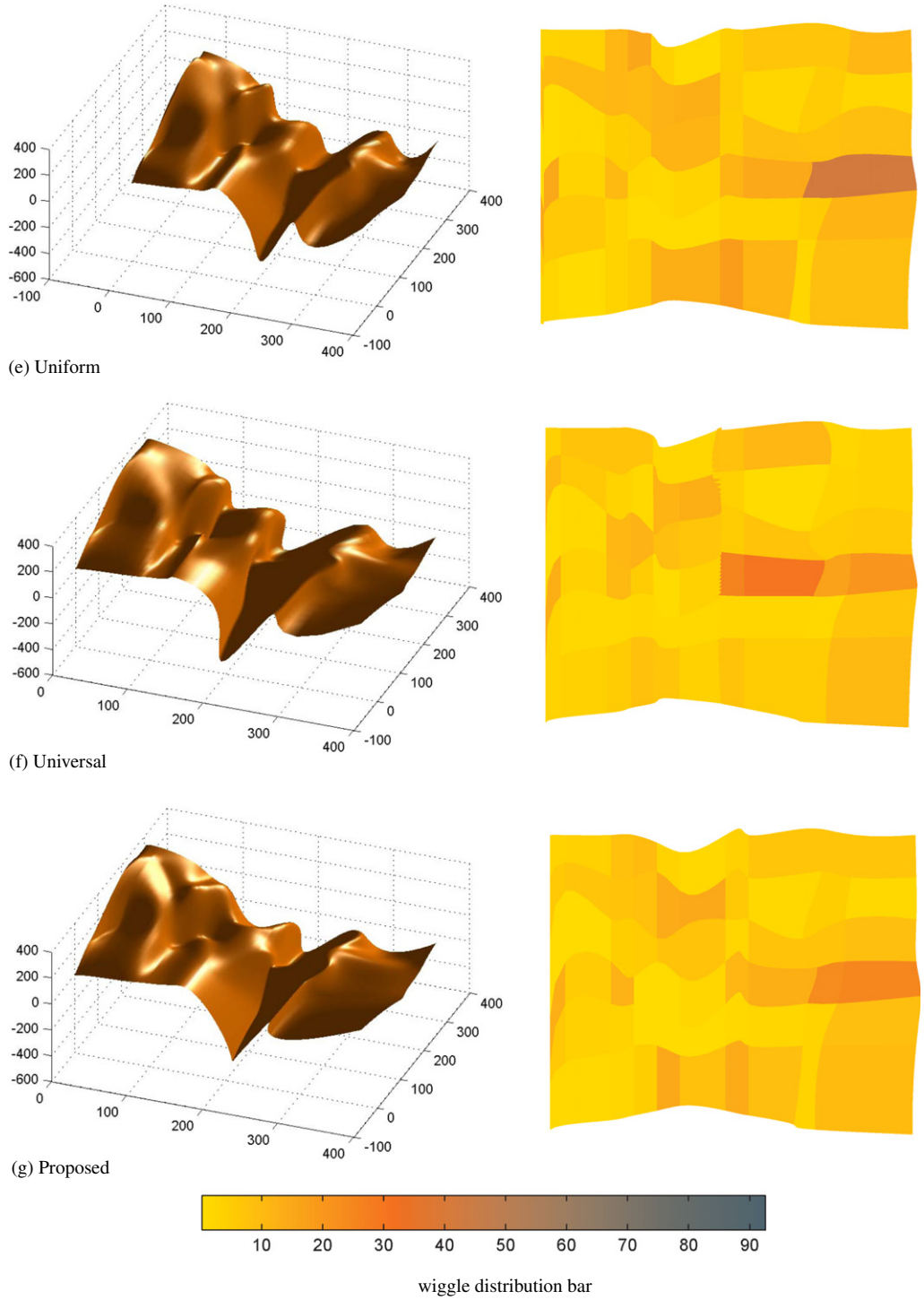


Fig. 21. (*continued*)

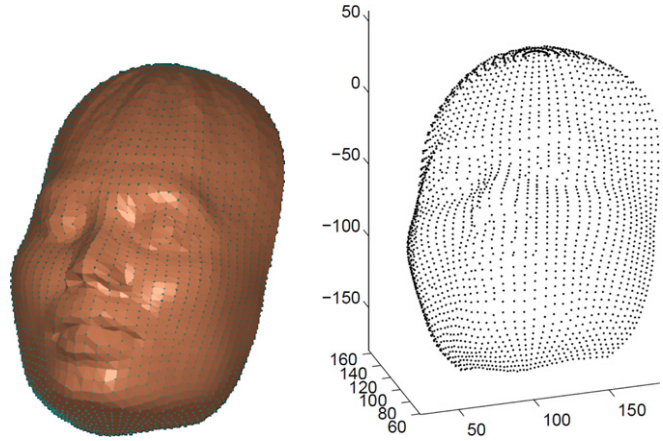
patch are demonstrated on the left and right in Fig. 21, respectively. For more clearly to verify the wiggle distribution of each method, $W_{\max}$ is listed to accompany W_s, where $W_{\max}$ is defined as

$$W_{\max} = \max_{0 \le i < n, 0 \le j < m} \{W_{i,j}\}.$$

From a comparison in Table 16, it is seen that the proposed method is best, then the centripetal method. Actually, these methods can be regarded as performing the same by their few wiggle differences, excluding the F-CHD method, which has an extraordinary large wiggle in one patch. The corresponding patch of significant wiggle of F-CHD is the dark part in Fig. 21(d).

In a free-form case, as shown in Fig. 22, is a 58 × 38 grids extracted from a scanned human face. Table 17 shows that the result of the proposed method outperforms all of other methods. In this case, all methods perform about the same results except the universal method. The rough surface in the left of Fig. 23(f) explains the greater wiggle value of the universal method, caused by the peaks.

These two results demonstrate the proposed method can perform in 3D space as well as in 2D plane. The properties of the proposed method are kept in surface interpolations, which satisfy the expectation with regard to the parameterization methods.



(a) Face surface. (b) Data points grid.

Fig. 22. Extracted human surface data grid.

Table 16
Wiggles of the results in Fig. 21.

Wiggle	Centripetal	Chord length	F-CEN	F-CHD	Uniform	Universal	Proposed
W_S	6.962490	7.189910	5.940640	8.039530	7.904030	6.389530	6.254370
$W_{\max}$	37.610400	35.416500	25.535400	92.648000	41.990600	30.643700	25.302200

Table 17
The wiggles of the resulting surfaces in Fig. 23.

Wiggle	Centripetal	Chord length	F-CEN	F-CHD	Uniform	Universal	Proposed
W_S	0.160468	0.161944	0.156754	0.158070	0.160880	1.733160	0.156365
$W_{\max}$	3.232640	7.396590	3.232640	3.756840	3.857050	1901.510000	5.110120

Table 18
Wiggles of the results in Fig. 27.

Wiggle	Centripetal	Chord length	Foley	F-CEN	F-CHD	Uniform	Universal	Proposed
W_C	20.7692	363.288	66.7336	77.8193	30.6381	8821.96	2.71709	4.06879

Table 19
Wiggles of the results in Fig. 28.

Wiggle	Centripetal	Chord length	Foley	F-CEN	F-CHD	Uniform	Universal	Proposed
W_C	194.796	407.514	3510530	225.635	319.11	138.427	27.6889	40.6887

5. Discussion

Form the studies of curve cases in Section 3 and surfaces cases in Section 4, the proposed method presents its fine properties over other methods. Among all the examples, loops or peaks were not occurred in the results of the proposed method, as well as the centripetal method. To more advantage, the new method significantly reduces the wiggle distance of the centripetal method, especially when interpolating an abrupt data distribution.

The experiments here are tested under degree 2–4, however, the higher degrees are used in some CAD applications. We test some of the examples in this paper with degree 6 and take brief comparisons to other methods. Fig. 24 shows the interpolating results on the data of Fig. 9 by the centripetal, proposed and F-CEN method in degree 6. These three methods perform finest while the degree changes from 2 to 4. As a result, F-CEN and the proposed method work more stable than the centripetal method. W_C of the centripetal, F-CEN and the proposed method are 131.969, 40.4499 and 43.7668, respectively.

Curves in Fig. 25 are the results that are interpolated by the centripetal, uniform and the proposed methods on the data of Fig. 12. The respective W_C of these three methods are 46.1017, 834.776 and 194.6. Although W_C of the proposed method is larger than that of the centripetal method, the interpolating curve of the centripetal has loops, and wanders far, as shown in Fig. 25. Therefore, the proposed method is preferred by concluding with Fig. 25 and W_C.

We also test the surface case of Fig. 22 by the centripetal, uniform and the proposed methods in degree 6. The results in Fig. 26 are similar to the previous test. But W_S of these three methods are 74.5536, 126.407 and 0.428188. It is seen that the proposed method are more stable than others under the degree changing. Nevertheless, the three outcomes are all accepted. In fact, the grids of this face are separated near uniformly. The

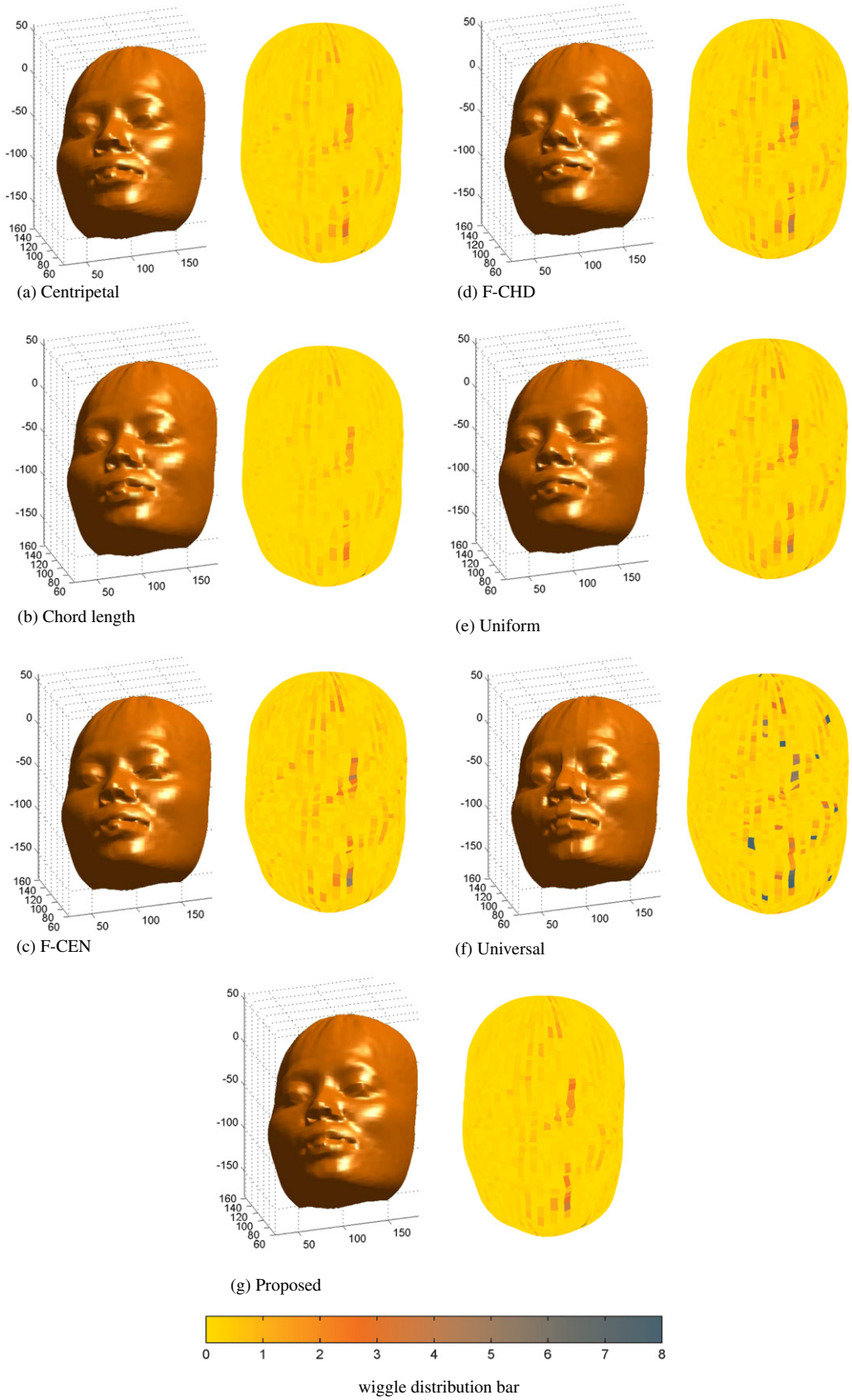


Fig. 23. The interpolating surfaces of Fig. 22 using different methods.

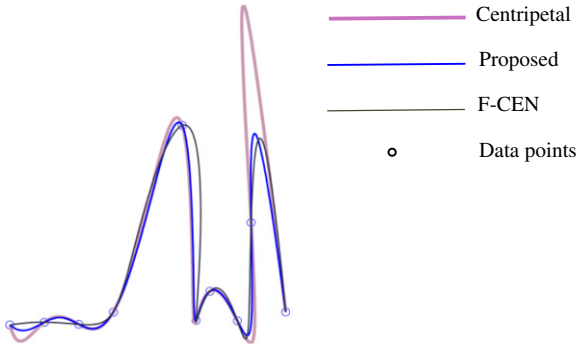


Fig. 24. Interpolating results of Fig. 9 by the centripetal, proposed and F-CEN method in degree 6.

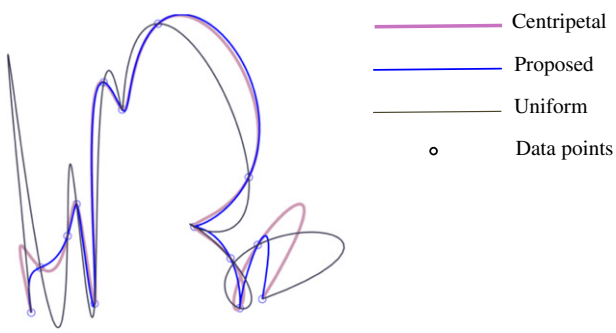


Fig. 25. Interpolating results of Fig. 12 by the centripetal, proposed and uniform method in degree 6.

interpolating results will be not too bad even there are points with high curvature.

On the other hand, the knot placement choosing also plays a critical role in curve and surface fitting. Uniform knot is one of the most popular determinations. However, it can result in a singular system of equations (Eq. (1)) [1]. Even so, we try some examples to see the difference of the results between the uniform and the averaging method of Eq. (4). Fig. 27 shows the results of cubic interpolation on the data of Fig. 7 by all methods with the uniform knot. The respective W_C are listed in Table 18. It is seen that, the results are worse than which with the averaging method except the universal method. Fig. 28 shows the results of cubic interpolation to the data of Fig. 9 except Foley, F-CHD, F-CEN for their very poor performances. Table 19 lists the corresponding wiggles of all methods. In fact, the universal is the best in these two cases. However, the wiggles of the proposed method are just a little larger than the universal method. It can be proven that the proposed method still works well with the uniform knot. Therefore, the proposed method shows its stability with both averaging and uniform knot vector. In advance, searching the optimal knot with the proposed method and other method is another subject.

Regarding the wide use of the approximation to fit a curve or a surface, we try the cubic B-spline approximation to the data in Fig. 22 by the simple least square method. The approximating result is shown in Fig. 29. The wiggle validation also works on approximation and the value is 11.1745. This result can be improved with adding constraints or other ways.

6. Conclusions and future work

In B-spline curve interpolation, node choosing plays an important role in determining the form of the resulting curve. Based on the centripetal method, we proposed a well-modified parameterization method which included the osculating circles of the data. The proposed method refines the interpolating results of the centripetal method, especially for the abrupt distribution of data. More advantages, the proposed method significantly outperforms other methods after comparing the wiggle evaluation, especially in the case of the abrupt-changing data. Even the wiggle value of the proposed method is not the smallest in some cases, the shape of the resulting curve is still acceptable. Furthermore, the drawbacks of interpolation, such as loops, peaks, or wiggles occurred less than in the other methods, regardless the data shape and interpolating degrees. However, the proposed method may not affect the centripetal method when the data is U-shaped.

The proposed method also has good performance on interpolating surfaces in free-form data types. Since the space curve interpolation is worked during the process of surface interpolation, it has been proven that the proposed method also works well with regard to space curve interpolation. According to these features, the proposed method was found to be much better and stable than other methods in curve and surface interpolation. Additionally, all results here verify that a cusp occurs easily in the resulting curves of the uniform and the universal methods.

Besides the proposed method, we determined a wiggle criterion function, to validate the performances of curve and surface interpolation, respectively. The wiggle validation computes both the distance deviation and the curvature which are most important features in the curve and surface fitting. These fine functions help us to clarify the level of fitness, no matter interpolation of approximation. We also evaluate the interpolation error to compare the wiggle validation. The two examples indicate that

(a) Centripetal. (b) Uniform. (c) Proposed.

Fig. 26. Interpolating results of Fig. 22 by the chord length, uniform and proposed method in degree 6.

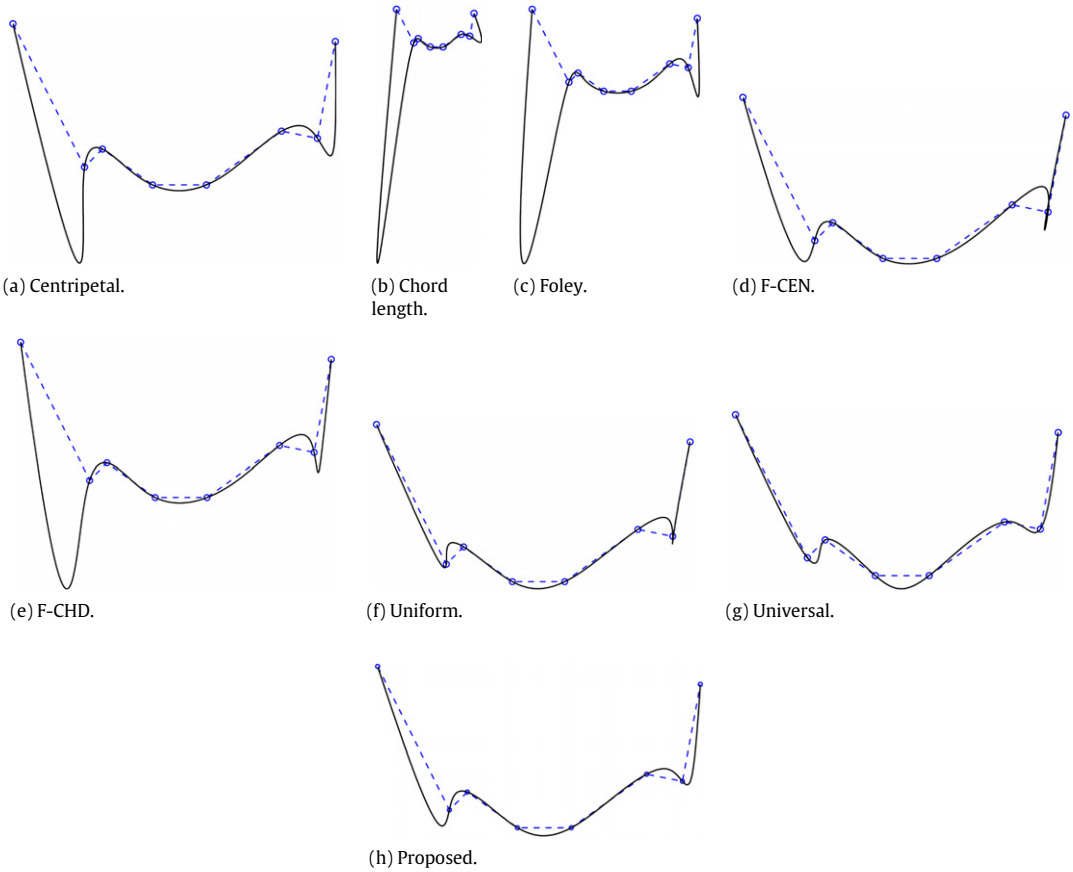


Fig. 27. Interpolating results of Fig. 7 by different methods with uniform knot.

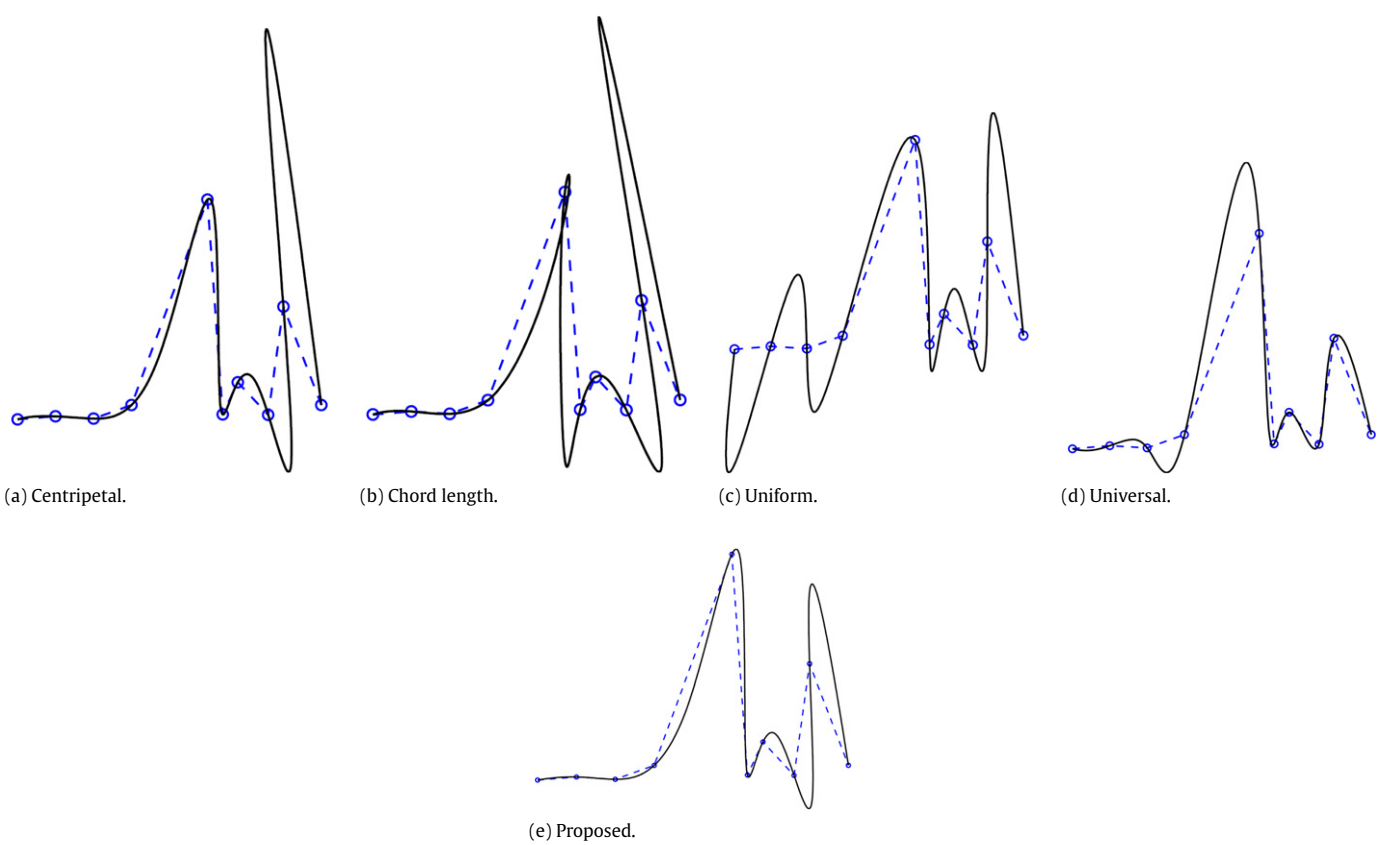


Fig. 28. Interpolating results of Fig. 9 by different methods with uniform knot.

Table A.1
Parameter of data points in Fig. 6.

Centripetal	0	0.21655	0.30266	0.43698	0.56240	0.72690	0.83031	1
Chord length	0	0.30328	0.35123	0.46790	0.56962	0.74463	0.81379	1
Foley	0	0.26709	0.33689	0.46306	0.55130	0.69805	0.80810	1
F-CEN	0	0.17405	0.26874	0.41566	0.53693	0.68544	0.84551	1
F-CHD	0	0.22012	0.28581	0.41701	0.53352	0.69590	0.83250	1
Uniform	0	0.14286	0.28571	0.42857	0.57143	0.71429	0.85714	1
Universal	0	0.09000	0.22125	0.40000	0.60000	0.77875	0.91000	1
Proposed	0	0.16869	0.26419	0.40882	0.56425	0.73610	0.85685	1

Table A.2
Parameter of data points in Fig. 8.

Centripetal	0	0.0756	0.1512	0.2291	0.4106	0.5906	0.6637	0.7456	0.8739	1
Chord length	0	0.0450	0.0899	0.1375	0.3964	0.6510	0.6930	0.7457	0.8751	1
Foley	0	0.0144	0.0323	0.0648	0.3045	0.5726	0.6424	0.7236	0.8795	1
F-CEN	0	0.0323	0.0706	0.1373	0.3175	0.5168	0.6277	0.7401	0.8933	1
F-CHD	0	0.0199	0.0435	0.0892	0.3349	0.5918	0.6647	0.7431	0.8901	1
Uniform	0	0.1111	0.2222	0.3333	0.4444	0.5556	0.6667	0.7778	0.8889	1
Universal	0	0.0643	0.1580	0.2857	0.4857	0.6286	0.7143	0.8420	0.9357	1
Proposed	0	0.0560	0.1353	0.2177	0.3993	0.5844	0.6712	0.7662	0.9005	1

Table A.3
Parameter of data points in Fig. 11.

Centripetal	0	0.0613	0.1196	0.1714	0.2621	0.3966	0.4483	0.5354	0.6559	0.7334	0.7960	0.8601	0.9334	1
Chord length	0	0.0443	0.0844	0.1160	0.2132	0.4264	0.4579	0.5474	0.7188	0.7896	0.8358	0.8844	0.9477	1
Foley	0	0.0232	0.0435	0.0815	0.1981	0.3616	0.4182	0.5265	0.6834	0.7771	0.8290	0.8805	0.9548	1
F-CEN	0	0.0303	0.0576	0.1105	0.2170	0.3429	0.4206	0.5222	0.6430	0.7348	0.7981	0.8616	0.9465	1
F-CHD	0	0.0230	0.0426	0.0803	0.1984	0.3638	0.4213	0.5288	0.6855	0.7763	0.8280	0.8801	0.9545	1
Uniform	0	0.0769	0.1538	0.2308	0.3077	0.3846	0.4615	0.5385	0.6154	0.6923	0.7692	0.8462	0.9231	1
Universal	0	0.0409	0.1006	0.2182	0.2727	0.3636	0.4545	0.5455	0.6364	0.7273	0.8182	0.8994	0.9591	1
Proposed	0	0.0399	0.0939	0.1472	0.2662	0.3990	0.4544	0.5545	0.6840	0.7711	0.8393	0.9023	0.9618	1

Table B.1
Wiggles of each segment in Fig. 7 with degree = 3.

Wiggle	Centripetal	Chord length	Foley	F-CEN	F-CHD	Uniform	Universal	Proposed
0	8.96283	42.2193	18.0063	5.55931	12.5886	1.95073	0.792848	5.15186
1	0.903199	0.574932	0.90538	0.48486	0.595983	1.69124	9.26081	0.47118
2	1.42883	2.45147	1.2912	1.67857	2.0909	1.08631	4.72019	1.92129
3	2.57856	2.307	1.56476	2.317	2.40409	3.18429	5.94917	3.58385
4	1.47496	2.87127	0.881825	0.351924	0.724031	0.693743	2.91331	1.68344
5	0.626194	0.253738	1.28204	3.91112	1.3828	2.67707	3.22711	0.518101
6	5.3433	11.0954	6.7319	2.55411	3.54736	2.47336	1.11245	3.18544
Maximum	8.96283	42.2193	18.0063	5.55931	12.5886	3.18429	9.26081	5.15186
Average	3.04541	8.82473	4.38049	2.40813	3.3334	1.96525	3.99655	2.35931

Fig. 29. Surface approximation of Fig. 22 with 26 × 17 control points.

the wiggle validation can display either distance deviation or unusual curvature changing. However, the wiggle values are used for the relative comparison. One cannot verify the level of the interpolating result only by its own wiggle value. Hence the details of the wiggle validation or its refinement, can be the subjects to investigate in the future.

Due to the fine advantages of our method, the proposed method can offer the good initial parameter values to do parameterization optimizations, representations or fitting optimizations.

Besides the node choosing, knot selection is the other critical issue in parameterization. The optimal knot placement conjoined with the proposed method can be discussed in future work. In addition, investigating on applying the proposed method to noisy or dense data on B-spline approximation or other parametric function will also be of interests in future research.

Appendix A

Tables of parameter values of each methods. (See Tables A.1–A.3.)

Appendix B

Table of wiggles of each segment in curves case. (See Tables B.1–B.4.)

Table B.2
Wiggles of each segment in Fig. 9 with degree = 3.

Wiggle	Centripetal	Chord length	Foley	F-CEN	F-CHD	Uniform	Universal	Proposed
0	1.585910	2.892280	1.391440	1.415530	1.435010	8.442040	0.084726	0.841376
1	0.243334	1.143970	0.270130	0.302920	0.279701	10.483400	2.274670	0.690776
2	7.556930	3.525590	7.125870	8.107920	7.946360	39.276800	19.227600	7.992470
3	11.005800	963.078000	53.613700	7.731250	36.832300	13.298100	33.251800	10.277300
4	16.390400	680.623000	70892.300000	23.823100	2559.280000	13.005200	3.364460	21.689200
5	4.538580	5.354310	5.817730	18.948600	6.620170	109.150000	4.287610	4.616240
6	4.653420	5.984930	4.652580	3.733480	3.634910	7.199510	23.923200	4.034560
7	13.569000	33.203300	18.293300	7.385040	16.735100	4.069210	5.533980	8.832080
8	60.309600	151.644000	36.081900	17.004100	29.358300	35.593800	13.095500	19.465000
Maximum	60.309600	963.078000	70892.300000	23.823100	2559.280000	109.150000	33.251800	21.689200
Average	13.317000	205.272000	7891.060000	9.827990	295.791000	26.724200	11.671500	8.715450

Table B.3
Wiggles of each segment in Fig. 11 with degree = 3.

Wiggle	Centripetal	Chord length	Foley	F-CEN	F-CHD	Uniform	Universal	Proposed
0	12.5595	14.1186	4.02462	4.0231	4.23284	12.0735	4.00644	5.1208
1	2.35074	0.437917	11.8303	5.99607	1.15394	10.6363	6.59526	0.875066
2	3.02034	12.7565	49.6286	35.639	6.28456	8.47218	6.70924	3.77266
3	8.23811	20.8279	7.79822	4.88509	7.95343	3.60791	6.22534	8.98497
4	3.29083	8.64163	4.79571	3.75051	1.3136	7.66013	8.6025	1.48176
5	1.67039	0.316492	0.487371	3.74076	1.04959	17.9736	30.9709	1.21367
6	0.325073	4.27419	3.09921	0.436437	1.22459	1.59095	1.09613	0.837795
7	2.7114	6.17046	3.61133	1.52549	2.95169	0.763751	0.803713	2.75626
Maximum	12.5595	20.8279	49.6286	35.639	7.95343	17.9736	30.9709	8.98497
Average	4.2708	8.44296	10.6594	7.49956	3.27053	7.84729	8.1262	3.13037

Table B.4
Wiggles of each segment in Fig. 12 with degree = 3.

Wiggle	Centripetal	Chord length	Foley	F-CEN	F-CHD	Uniform	Universal	Proposed
0	16.06980	16.54980	13.21050	12.73750	13.70990	17.23340	7.19127	8.31851
1	5.66250	4.26787	1.64820	1.61461	1.59193	11.37430	2.20758	1.82738
2	3.23369	2.31106	15.10180	16.91730	15.88680	82.70520	6435.73000	3.24040
3	6.06807	39.91500	41.23500	24.38940	46.59810	23.24480	5.60594	20.70460
4	40.93780	5509.16000	27.02390	10.45860	30.49370	2.41924	15.89740	17.38490
5	2.61428	2.04904	1.93226	9.27280	2.23801	48.27100	72.40170	2.37138
6	10.20620	28.16350	14.85980	22.73650	13.91230	34.70700	43.74740	12.62370
7	75.24150	155.75400	95.18640	59.79180	97.41820	36.86400	37.91140	71.72130
8	11.57480	17.70870	19.63320	15.68450	19.02390	42.09540	41.47350	15.98020
9	5.21420	5.64174	7.75532	6.35339	7.23940	8.59202	193.02900	12.90050
10	32.60970	7.85058	5.91016	8.37832	6.31008	81.11120	57.51940	12.02990
11	6.56590	9.42293	9.63287	10.94640	9.52090	1.09892	3.78504	3.05385
12	43.21980	37.91370	14.90950	15.27660	15.07130	48.06160	12.12100	12.36400
Maximum	75.24150	5509.16000	95.18640	59.79180	97.41820	82.70520	6435.73000	71.72130
Average	19.93990	448.97700	20.61840	16.50440	21.46260	33.67520	532.97100	14.96310

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